

LEAP: Trajectory-Level Evaluation of LLMs in Iterative Scientific Design

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LLMs are increasingly deployed in autonomous laboratories, under the assumption that their domain priors and reasoning over iterative feedback let them converge on good designs in fewer iterations than feedback-only baselines. Current iterative scientific design benchmarks, however, score only outcome snapshots at fixed horizons. This leaves the learning trajectory unmeasured, even though the trajectory is what captures learning efficiency, where each iteration saved is a real saving in cost and time. Motivated by this, we examine three evaluation choices that change the conclusions one draws about LLM learning efficiency in iterative scientific design: what to measure, what baseline to compare against, and what to ground against. We introduce **LEAPBENCH**, Learning Efficiency in Adaptive Processes, a 55-task framework that pairs a best-so-far area under the curve (AUC) trajectory metric with a classical Bayesian-optimization reference and an audit grounded in published literature. Applied to eight contemporary LLMs, switching from final-outcome to trajectory scoring changes the best-model decision on 53% of tasks at matched horizons, and exposes efficiency gains overlooked by outcome-based scoring. LLMs do not outperform a classical Bayesian baseline. On 16 biology tasks where the oracle’s reward signal is aligned with configurations from the published-best design, domain-aware prompting leads to LLM choices that match the published-best’s ≈ 10 percentage points less often than domain-agnostic prompting at iteration 30. The pattern is sharpest on 6 tasks where the literature-typical and published-best configurations diverge, and domain-agnostic prompting matches the published-best more often on all 6. The trajectory metric also doubles as a tractable training target. Offline reinforcement learning with the metric as a reward improves performance on 14 of 21 held-out tasks.

Keywords: AI for science, in-context learning, LLMs as optimizers

1 Introduction

Scientific discovery is iterative, and each iteration costs time and money. LLMs’ exposure to domain information in training (domain priors) and their ability to reason over iterative feedback (Brown et al., 2020) are assumed to help find optimal results within fewer iterations, and as such, they are currently being deployed inside autonomous-lab loops (Boiko et al., 2023; FutureHouse, 2025; Smith et al., 2026) or are an active research target (Anthropic, 2025; Gottweis et al., 2025; Nguyen and Grover, 2025; van Dijk and Vrkic, 2026).

Current iterative scientific benchmarks focus on final outcomes and leave this learning efficiency assumption largely untested. Static-correctness benchmarks grade single-shot answers against a known target (Mirza et al., 2025; Rein et al., 2024; Wang et al., 2024). Single-shot experimental-design benchmarks score a proposed design at one timestep (Guo et al., 2025; Liu et al., 2025; Majumder

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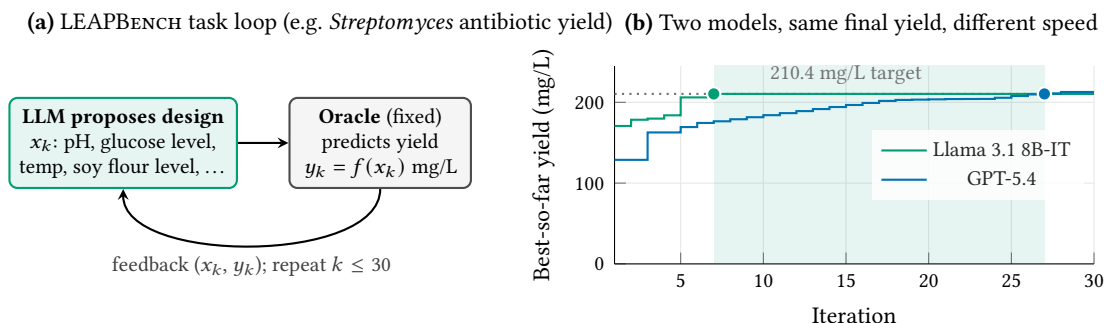


Figure 1. Outcome-only evaluation calls two models tied on a task where one reaches the target twenty iterations earlier. (a) Each LEAPBENCH task is an iterative loop: the model proposes a parameter vector (e.g., pH, glucose concentration, temperature on a *Streptomyces* synthesis task), receives a scalar yield from a fixed surrogate oracle, and uses prior feedback for the next proposal, for up to 30 iterations. (b) Both models reach the 210.4 mg/L target, but Llama 3.1 8B-IT reaches it 20 iterations earlier than GPT-5.4 (iteration 7 vs. 27). Outcome-only evaluation calls them tied; bsf-AUC ranks Llama first.

et al., 2025; Shojaee et al., 2025). Sequential LLM-as-optimizer studies run multiple rounds but evaluate each run on its final outcome (Madaan et al., 2023; Nguyen and Grover, 2025; van Dijk and Vrkic, 2026; Yang et al., 2024), not the trajectory of proposed designs leading to it. Yet how a model gets to its final score matters. Figure 1 shows two LLMs reaching the same final yield on a *Streptomyces* antibiotic synthesis task, one twenty iterations ahead of the other. One LLM trajectory is more efficient in scientific iteration, but endpoint metrics would label them tied. In autonomous-lab settings, iterations saved translate to real savings in time and money (Boiko et al., 2023; FutureHouse, 2025).

We focus on parameterized iterative scientific design, one of the settings where recent AI-for-science deployments operate, such as LLM-driven cell-free protein synthesis cost reduction (Smith et al., 2026). When LLMs perform scientific optimization tasks, how can we tell whether they learn efficiently, and when do domain priors help with learning efficiency? Answering this requires unpacking three evaluation choices that current iterative-LLM benchmarks settle implicitly: what trajectory property to measure, what reference point to compare against, and whether trajectory choices are scored against the surrogate oracle alone or also audited against published best-performing configurations observed in the literature.

We introduce LEAPBENCH, Learning Efficiency in Adaptive Processes, a 55-task evaluation framework that operationalizes all three axes. We use best-so-far AUC (bsf-AUC@k) as the trajectory metric, Gaussian-process Upper Confidence Bound (GP-UCB) as the classical sequential-search baseline, and an audit grounded in published wet-lab outcomes on tasks, including where literature-typical and empirical-best diverge. A paired prompt manipulation isolates the joint effect of any domain priors and any domain-specific feedback responses the model brings to the task by removing all semantic context in the domain-agnostic prompt, though we do not isolate a single causal mechanism underlying the observed differences between prompt conditions.

The 55 tasks aggregate 2,719 experimental observations drawn from 375 published biology studies and 10 peer-reviewed randomized controlled trials (RCTs). Many studies contribute multiple datapoints across or within tasks. LEAPBENCH does not address free-form hypothesis generation

or scientific reasoning.

Applying the LEAPBENCH framework to eight frontier LLMs, we demonstrate for each axis:

1. *What we measure matters.* On biology, bsf-Outcome@30 and bsf-AUC@30 **rank different best models on 24 of 45 tasks**, with 12 of those 24 being rank-3-or-deeper reorderings (§4.1). Outcome-based scoring overlooks learning efficiency gains that bsf-AUC does not. On 14 of these 24 tasks, the bsf-AUC winner and the bsf-Outcome winner converge to within 1% of each other by iter 30 yet the bsf-AUC winner arrives 8 iterations earlier (median iteration 7 vs 15). As a trajectory-level metric, bsf-AUC also doubles as a tractable training target. Offline GRPO with a bsf-AUC reward improves performance on 14 of 21 held-out tasks and transfers to never-seen education tasks (§4.4).
2. *What we compare against matters.* Moving beyond LLM-to-LLM comparison, we find that **LLMs do not outperform GP-UCB** under matched-information conditions. The prompt manipulation moves performance within the GP-parity band, not across it (§4.2).
3. *What we ground against matters.* Grounding LLM trajectory choices against published wet-lab outcomes, rather than against the surrogate oracle alone, reveals that domain-aware prompting underperforms domain-agnostic on biology. On 16 biology tasks where the oracle’s reward signal is aligned with configurations from the published-best design, domain-aware prompting leads to LLM choices that match the published-best’s ≈ 10 percentage points less often than domain-agnostic prompting at iter 30. On the 6 tasks where the literature-typical answer differs from the published best, **domain-agnostic prompting finds the published best more often than domain-aware prompting on all 6** (§4.3). Yet domain-aware prompting helps on education tasks.

Together, these findings show that evaluation design choices materially change conclusions about LLM learning efficiency. Practitioners deploying domain-aware prompts in autonomous-lab systems may want to test domain-aware vs. domain-agnostic prompting on representative tasks before committing, particularly where the literature-typical answer may differ from the experimental optimum.

2 Related work

LLM Benchmarks for Scientific Tasks. Existing scientific LLM benchmarks score the *endpoint* of scientific work: final-answer correctness on static problem suites (Laurent et al., 2024; Mirza et al., 2025; Rein et al., 2024; Wang et al., 2024), final-design quality on inverse-design and discovery benchmarks (Chen et al., 2025; Guo et al., 2025; Liu et al., 2025; Majumder et al., 2025; Mitchener et al., 2025; Shojaee et al., 2025), and final-outcome scores on sequential LLM-as-optimizer studies (Madaan et al., 2023; Nguyen and Grover, 2025; van Dijk and Vrkic, 2026; Yang et al., 2024) and long-horizon agent simulations (He et al., 2026). Trajectory-level and sample-efficiency metrics appear in games (Chollet, 2019; Kaiser et al., 2020; Mnih et al., 2015; Silver et al., 2016) (episodes-to-mastery), continual learning (Asawa et al., 2026) (across-instance learning gain in software engineering and data science), and intermediate-reasoning evaluation (Lightman et al., 2024; Uesato et al., 2022; Zheng et al., 2025) (per-step correctness within a single response). What is missing is a metric for how efficiently a model converts iterative oracle feedback into improvement on a single sequential

design task. How efficiently an LLM converts iterative feedback into improvement on a scientific task remains unmeasured.

Rank Disagreement across Metrics. Prior work has surfaced task- and metric-level rank disagreement in AutoML (Gijbbers et al., 2024), deployment fairness (Schrouff et al., 2022), and LLM evaluation (Liang et al., 2023; Sclar et al., 2024). These studies examine static, single-shot evaluation. LEAPBENCH surfaces analogous disagreement on the same trajectory across three axes: process-vs-outcome metric, baseline (LLM-only vs. classical Bayesian optimization), and grounding (surrogate oracle vs. published wet-lab outcomes).

Domain-Knowledge Effects in LLM Reasoning. LLMs default to learned patterns on counterfactual tasks (Wu et al., 2024) and show anchoring and default-bias behaviors in static prompts (Jones and Steinhardt, 2022; Suri et al., 2024), echoing the human anchoring bias (Tversky and Kahneman, 1974). Whether similar patterns persist under iterative experimental feedback has not been tested.

LLMs as Iterative Optimizers. Gaussian-process Bayesian optimization (BO) (Shahriari et al., 2016; Srinivas et al., 2010) is a reproducible, sample-efficient reference for black-box iterative search. Existing LLM-as-optimizer work (Madaan et al., 2023; Nguyen and Grover, 2025; Yang et al., 2024) reports final outcomes without trajectory-level metrics or domain-knowledge stratification. Reinforcement learning for experimental design (van Dijk and Vrkic, 2026) reports success rates at fixed horizons rather than trajectory-integrated metrics, and lacks a classical-BO reference for normalization. LEAPBENCH addresses both gaps.

3 LEAPBENCH: tasks, metrics, and evaluation protocol

Each LEAPBENCH task is an iterative scientific design loop. The model proposes an experimental design, receives a scalar outcome from a fixed oracle, and proposes the next.

Learning efficiency metric (bsf-AUC). Learning efficiency captures whether each iteration of feedback is converted into improvement. We operationalize this as the area under the best-so-far curve at iteration k (**bsf-AUC@k**), capping each run at 30 iterations. bsf-AUC integrates the running maximum, rewarding both how high the best-so-far climbs and how quickly, so a trajectory that re-asserts the same proposal despite feedback signaling otherwise accumulates bsf-AUC slowly even when its final value matches a faster learner’s. Alternative sample-efficiency metrics are less appropriate fits for this setting. Cumulative regret penalizes exploration that doesn’t immediately pay off, a poor fit when finding a single best candidate is the goal. Iterations-to-target requires a task-specific threshold whose choice itself changes rankings. Time-to-best is censored at the budget horizon and can rank a model that plateaus early above one that reaches a higher value later.

3.1 Tasks: 45 biology, 10 education

LEAPBENCH comprises 55 sequential tasks. Each task is one optimization problem over a 3–12 dimensional parameter space, with oracles trained from data aggregated from published studies reporting comparable (parameters, outcome) data. The 45 biology tasks aggregate 2,719 experimental datapoints from 375 published studies, ranging from microalgae cultivation to mammalian cell engineering to enzymatic biocatalysis (full task list in §A.2). Many studies contribute multiple

datapoints or appear in multiple tasks. The 10 education tasks are derived from peer-reviewed RCTs whose known correct mechanisms serve as a validation set; education-panel companion analyses are in the appendix (§A.10, §A.26, §A.16). For each task, we train a supervised regression model on the published literature dataset and use it as the oracle (leave-one-out [LOO] R^2 median 0.89 biology / 0.32 education). The literature-typical and best-result references used in the audit (§4.3) are computed from the published literature dataset itself, not from the oracle. Models iterate over 30 rounds with 4 runs per (model, task, domain-awareness condition) combination. All tasks, oracles, prompts, and baselines are released.

3.2 A paired domain-knowledge intervention

Each model is evaluated under two conditions that share an identical oracle and parameter space. In **domain-aware prompting**, the prompt names the domain (e.g., “CHO antibody expression titer”) and parameters keep real names with units and named categoricals (e.g., cell_line: HEK293T, HEK293, ...). In **domain-agnostic prompting**, the prompt is generic (“optimize a black-box function”) and parameters are renamed ($X_1, X_2, \dots$; $C_1: A, B, \dots$), preserving ranges and cardinalities but stripping semantic content.

The oracle, GP-UCB, and domain-agnostic prompted LLMs all operate on the same semantics-free substrate. The oracle is trained on numeric features scaled to the unit interval plus one-hot categoricals, with no domain names, parameter units, or semantic labels. Domain-aware prompting is the only condition that adds semantic framing, so its contrast with domain-agnostic targets the joint effect of any domain priors the model brings and any domain-specific feedback handling, with the oracle, parameter space, and GP-UCB reference held fixed. Full templates in §A.3.

3.3 GP-UCB as a classical baseline

Each model run is paired with a GP-UCB baseline (Shahriari et al., 2016; Srinivas et al., 2010) (Matérn 5/2 kernel, $\beta = 2.0$, 200 runs per task, 1,000 for 4 minimization tasks with higher initial variance). GP-UCB fits a Gaussian process to observed data to estimate both the expected value of the objective and its uncertainty, then selects the next query by maximizing an upper confidence bound on the objective. This amounts to choosing points that are either already promising or highly uncertain, providing a principled trade-off between exploitation and exploration. GP-UCB is a standard, sample-efficient classical baseline for sequential search; we use it as a common normalization for cross-LLM comparison. Robustness against an alternative BO baseline, Heteroscedastic Evolutionary BO (HEBO) (Cowen-Rivers et al., 2022), is in §A.4.

3.4 Metrics: trajectory, baseline comparison, and published wet-lab grounding

Metrics group by the evaluation axis they operationalize. Operational definitions are in §A and statistical conventions in §A.5.

Trajectory metrics (axis: what to measure). The trajectory metric *bsf-AUC@k* (area under the best-so-far curve over k iterations) and the outcome metric *bsf-Outcome@k* (the best-so-far value at iteration k , equivalently the running maximum at the end of the trajectory of length k) separate fast-learning ability from eventual peak. *GP-normalized bsf-AUC* expresses each LLM’s

bsf-AUC as the relative gap to a task-specific GP-UCB baseline (0 = GP parity); rankings are robust to normalization choice.

Baseline comparison (axis: what to compare against). The *proportion of tasks outperforming GP-UCB* at a given horizon is the fraction where the model’s median run strictly outperforms GP-UCB’s median, with 50% as the null in expectation. The *domain-aware win rate* is the fraction of (model, task) pairs where the domain-aware run outperforms the matched domain-agnostic run on bsf-AUC@30.

Wet-lab grounding (axis: what to ground against). To detect when LLMs follow literature priors at the cost of finding the empirical optimum, an audit (§4.3) scores each iteration’s proposed design against the **published-best design**, the design with the highest measured target value in the source literature, rather than against the surrogate oracle alone. The headline metric is the *LLM-best-result match rate*: the fraction of iterations whose proposal matches the published-best design’s value on the task’s key categorical column. The audit is run on two nested subsets: (a) the 16 biology tasks where the surrogate oracle’s predicted-best aligns with the published-best so feedback can in principle distinguish it, and (b) the 6 of those tasks where the *literature-typical value*, the most-frequent categorical in the source-literature dataset, differs from the published best so finding the empirical peak requires exploring beyond literature consensus. A blinded *biology-expert review* (§C) provides supporting qualitative evidence.

3.5 Experimental setup

We evaluate eight LLMs spanning frontier and small-model tiers (Claude Opus 4.7, Sonnet 4.6, GPT-5.4, GPT-5.4 Mini, Gemini 3.1 Pro, 3 Flash, DeepSeek V3.2, Llama 3.1 8B Instruct (Anthropic, 2026a,b; DeepSeek-AI, 2025; Google DeepMind, 2026a,b; Grattafiori et al., 2024; OpenAI, 2026a,b)) on all 55 tasks under both prompt conditions, with 4 independent runs per (model, task, condition) cell (inference details in §A.7). All LLM API calls run with web search disabled, so any domain-prior contribution is attributable to training rather than retrieval-time lookup. Claude Opus 4.7’s safety filter refused 9 high- R^2 biology tasks (§A.8), leaving 46 with usable trajectories. Panel-level results are robust to its exclusion (§A.6). Selection-bias caveats are in §5.

4 Evaluation choices change conclusions about learning efficiency

4.1 Trajectory metrics and outcome metrics disagree on the best model, and outcome metrics overlook iterations saved

We first ask whether the choice between trajectory scoring (bsf-AUC) and outcome scoring (bsf-Outcome) changes the best model identified on the same trajectories. Across the 45 biology tasks, learning-efficiency (bsf-AUC@30) and final-outcome (bsf-Outcome@30) pick a different top-ranked model on 24 of them at matched horizons (53%, Figure 2a). Of the 24 disagreements, 12 are close rank-1-vs-2 swaps where the top two models tie under one metric, and 12 are rank-3-or-deeper reorderings, accounting for 27% of all 45 biology tasks, where the model picked under one metric is not even among the top two under the other (§A.11). Overall rankings remain positively correlated, so the disagreement is local rather than uniform across the leaderboard. The disagreement is also stable across horizons. On the 45 biology tasks the tie-aware-strict bsf-AUC@ k vs. bsf-Outcome@ k

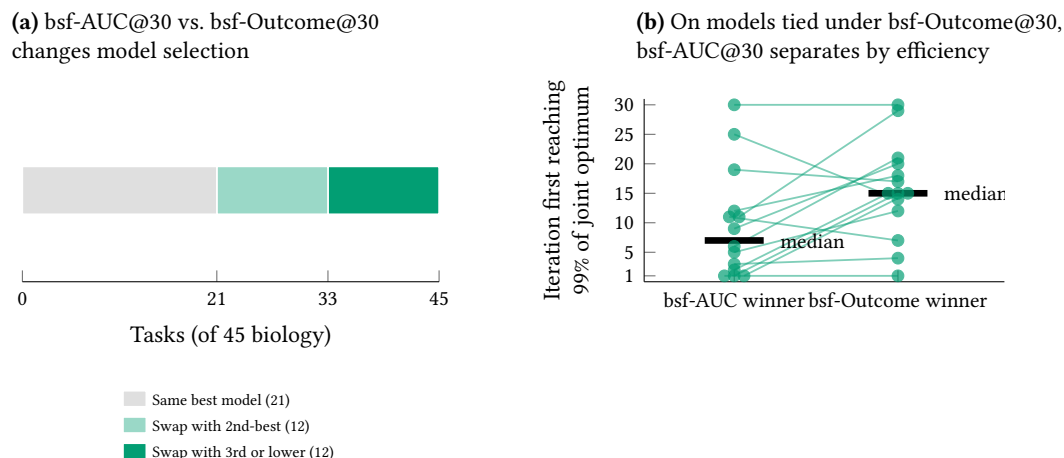


Figure 2. bsf-AUC@30 and bsf-Outcome@30 pick different best models, and bsf-Outcome doesn’t distinguish more efficient models. (a) Per-task: bsf-AUC@30 and bsf-Outcome@30 pick different top models on 24 of 45 biology tasks (53%), split between rank-1-vs-2 swaps (12) and rank-3-or-deeper reorderings (12). (b) On the 14 convergent disagreements (winners within 1% at iter 30), each line is one task showing the iteration each winner first reaches 99% of the shared optimum; black bars mark cross-task medians. Stats and details in §4.1.

disagreement rate ranges from 36% (16 of 45) at $k = 5$ to 53% (24 of 45) at $k = 30$, and per-model rank vs. GP-UCB shifts non-trivially across horizons (§A.9; DeepSeek catches up, Gemini 3 Flash drifts down, GPT-5.4 peaks early then fades). Horizon choice is therefore not the lever that resolves the metric disagreement.

bsf-Outcome scoring loses trajectory information that bsf-AUC captures. On 14 of the 24 biology disagreement tasks, the bsf-AUC winner and the bsf-Outcome winner end within 1% of each other at iter 30, and the count of convergent tasks rises monotonically as that gap threshold widens (§A.1). On the other 10 tasks the bsf-Outcome winner ends meaningfully higher, so the two winners do not share a common iter-30 target and time-to-peak is not comparable. We restrict to the 14 convergent cases because both winners reach the same final neighborhood there, isolating speed from final score. Figure 2b reports the iteration at which each winner’s median best-so-far curve first reaches 99% of the joint iter-30 optimum. The bsf-AUC winner’s cross-task median is iter 7 and the bsf-Outcome winner’s is iter 15 (paired Wilcoxon $p = 0.014$). The bsf-AUC winner reaches the threshold strictly earlier on 9 of 14 tasks, ties on 2, and reaches it later on 3. Under bsf-Outcome scoring these models look tied. Under bsf-AUC scoring they are eight iterations apart. This is the efficiency-gap pattern that motivates Figure 1.

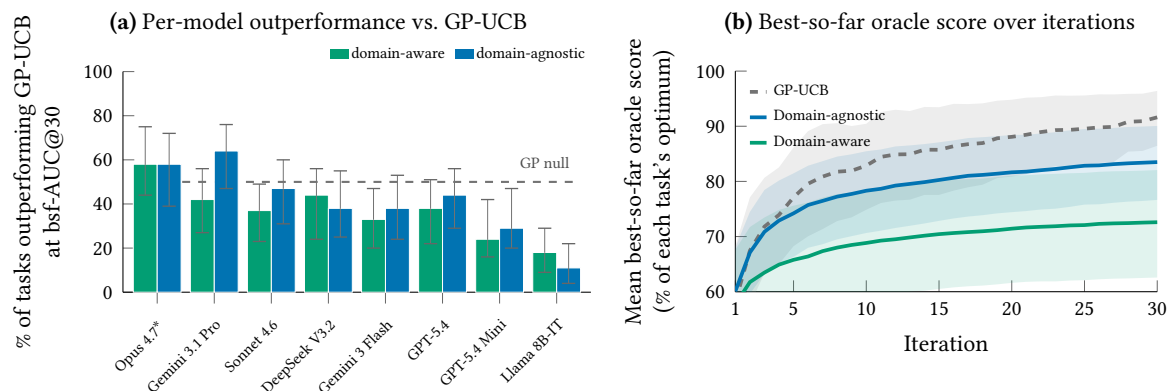


Figure 3. No LLM outperforms GP-UCB on biology bsf-AUC@30, though the trajectories trace different paths to the same neighborhood. (a) Per-model fraction of 45 biology tasks where the model’s median bsf-AUC@30 outperforms GP-UCB’s, under domain-aware (teal) and domain-agnostic (cobalt). Dashed line: 50% null. Error bars: 2-level bootstrap 95% CIs (§A.5). (b) Task-averaged best-so-far trajectory, normalized to each task’s optimum (1.0 = optimum). Asterisk on Opus 4.7’s bar marks 36-of-45 task coverage.

4.2 LLMs do not outperform classical Bayesian optimization

Most LLM-vs-LLM benchmarks compare model pass rates against each other, but contextualizing LLMs against an external classical reference is what tells us whether semantic priors and reasoning over feedback give LLMs the edge that current autonomous-lab deployments (Boiko et al., 2023; FutureHouse, 2025; Smith et al., 2026) bet on. We ground each LLM run against GP-UCB on the same oracle and parameter space (no semantic information), and ask whether LLMs outperform a classical baseline given these prior advantages.

LLMs do not outperform GP-UCB on biology bsf-AUC@30 under either prompt condition. No model’s 95% CI clears the 50% null under the 4-run-matched bootstrap (Figure 3a). The highest pass rate under domain-agnostic is Gemini 3.1 Pro at 64% [47, 76], and the lowest is Llama 3.1 8B at 11% [4, 22]. Adding semantic context with domain-aware prompts does not change the conclusion. Claude Opus 4.7 reaches 58% [44, 75] on its 36 covered biology tasks, with worst-case extrapolation accounting for 9 high- R^2 task refusals giving $\approx 47\%$ on the full panel. We sampled GPT-5.5 on the 5 closest-to-GP-UCB biology tasks, and it cleared GP on 1 of 5 (§A.13). Figure 3b shows the trajectories trace different paths to the same neighborhood, with GP-UCB $\approx$ domain-agnostic $>$ domain-aware in point estimates throughout iters 1–30 and task-clustered 95% CI bands that overlap. Wide CIs reflect the conservative 2-level bootstrap, which resamples tasks plus LLM runs plus GP runs. The per-task pair-level comparison between prompt conditions is in §4.3. The per-model conclusion is robust to baseline choice: HEBO (Cowen-Rivers et al., 2022) gives the same per-model pattern (§A.4), and a 200-run GP baseline keeps point estimates within 8 percentage points and preserves ordering (§A.12).

4.3 A published wet-lab audit: how domain priors impact iterative learning

Domain priors are often assumed to improve iterative optimization performance in autonomous-lab settings via LLMs. Domain-aware prompting does help on the education validation set, with a 73% pair-win at iter 30 versus 42% on biology (§A.6). The rest of this section focuses on the

biology side of this asymmetry, where domain-aware prompting underperforms domain-agnostic. Domain-awareness may influence two components of bsf-AUC: *initial proposal quality* at iter 1 and *subsequent adaptation to feedback* from iter 1 to iter 30. We use domain-agnostic prompting as a control for LLM behavior without domain priors, then test both mechanisms broadly on biology in Figure 4 and focus on what happens with literature-typical answers in Figure 5. Both audits anchor against the published-best datapoint for a task, the design with the highest measured target value in the source literature, rather than the surrogate oracle.

On the 16 *feedback-actionable* biology tasks, where the surrogate oracle’s predicted-best is within 5% of the published-best so feedback can in principle distinguish it, domain-aware prompting matches the published best 10.4 percentage points less often than domain-agnostic at iter 30 (mixed-effects linear probability model, one-sided $p = 0.004$). For each task, we score the LLM’s choice on a single *key categorical* column, the column whose oracle-score spread across observed values is largest. The threshold is set a priori, and the result is robust at ≥ 0.90 , ≥ 0.95 , and ≥ 0.99 alignment levels (all one-sided $p < 0.005$, §A.15). Iter-1 published-best match rates are indistinguishable between conditions (coef +0.01, $p = 0.60$), even though iter-1 oracle scores under domain-aware are 3.78 percentage points higher (priors do raise the starting score; see §4.2 for iter-30 oracle behavior). The iter-1 to iter-30 climb in match rate is 11.3 percentage points smaller under domain-aware ($p = 0.004$). Parameter-space diversity is in fact higher under domain-aware (§A.14), so this is not under-exploration. 7 of 8 LLMs match the published best less often under domain-aware than under domain-agnostic at iter 30, with Claude Opus 4.7 the exception (+2.08 percentage points on its 36-task covered subset). **This pattern points to a feedback-responsiveness gap under domain-aware, not an initial-quality one.**

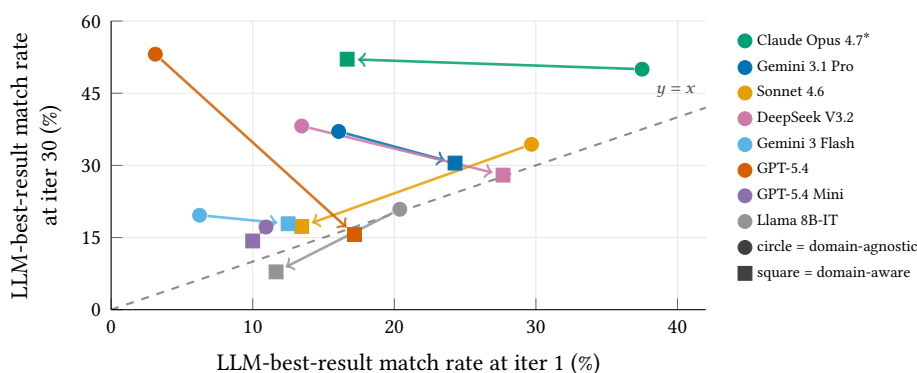


Figure 4. On biology tasks where feedback is actionable, domain-aware prompting reduces the published-best match rate at iter 30 by ~ 10 percentage points. Per-model LLM-best-result match rate at iter 1 (x) vs. iter 30 (y) on 16 feedback-actionable biology tasks (definition in §4.3). Each model has a circle (domain-agnostic) and a square (domain-aware), connected by an arrow from domain-agnostic to domain-aware. Dashed $y = x$ line marks no change. *Opus 4.7’s 9 task refusals reduce its domain-aware coverage; it is the per-model exception.

Focusing on the 6 biology tasks where the literature-typical answer for the key categorical column differs from the published best sharpens the failure (full criteria, exclusions, and threshold robustness in §A.17). On these 6 tasks, domain-agnostic selects the published best on 36.1% of iterations versus 20.7% for domain-aware (paired Wilcoxon $p = 0.031$, 6 of 6 tasks reversed). Six

of eight LLMs show this direction. The pattern is robust to gap-threshold variants and to adding GPT-5.5 (§A.13). Figure 5b shows the microalgae instance. Domain-aware disproportionately selects photobioreactor, the literature-typical reactor type, on 57.5% of iterations and reaches the published-best raceway pond on only 6.7%, while domain-agnostic explores its way to raceway pond on 84.2% of iterations. On 4 of 6 literature-divergent tasks, the oracle’s reward for the published best exceeds its reward for every more-modal literature value (§A.19), so feedback can in principle distinguish it. Domain-agnostic finds it more often despite identical oracle feedback, indicating the asymmetry is prompt-driven rather than a scoring artifact. On education, where literature consensus from RCT meta-analyses tends to coincide with the published best, the same audit shows domain-aware finds the correct mechanism more often than domain-agnostic (§A.16).

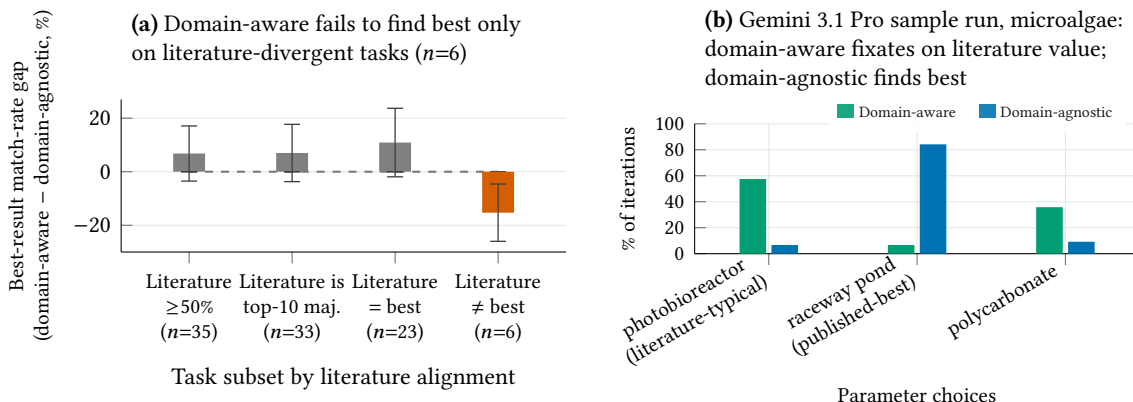


Figure 5. When finding the published best requires exploring beyond the literature-typical answer, domain-aware prompting selects it less often than domain-agnostic on all 6 biology tasks. (a) Best-result match-rate gap (domain-aware – domain-agnostic) with 95% CI across four nested subsets; only the literature-divergent subset (where finding the optimum requires exploring beyond the most-frequent literature value) clears zero negative. (b) Anchoring instance: Gemini 3.1 Pro on the algae biomass task. Raceway pond is the published-data optimum at 1.13 g/L (Yang et al., 2018), but photobioreactor is the standard literature reactor. Domain-aware fixates on photobioreactor; domain-agnostic explores its way to raceway pond. Iteration-level rates in §4.3.

Search is deliberately disabled at inference for both audits, both to isolate the prior contribution from runtime retrieval and to mimic deployment conditions where the novel designs the LLM proposes will not yet be in the published literature. The failure persists when the source paper is published before the LLM’s likely training data cutoff period (§A.20). Forcing web search on for Claude Opus 4.7 does not rescue domain-aware on the 6 literature-divergent tasks or on Opus’s 5 worst-domain-aware feedback-actionable tasks (§A.21). The 6/6 literature-divergent reversal is also direction-robust to three prompt-template perturbations (§A.18). A blinded review with $N=3$ biology experts, primed with a definition of literature-prior persistence, classified trajectory patterns consistent with the audit’s findings (§C). The priming precludes this from being independent confirmation, but it provides supporting qualitative evidence.

4.4 The trajectory metric is also a tractable training target

A trajectory-level evaluation metric is most useful if it can also be optimized as a training target. We test whether bsf-AUC@k responds to reinforcement learning, since most current RL post-training

rewards single-turn correctness (Ouyang et al., 2022; Shao et al., 2024) rather than multi-iteration improvement.

We LoRA-fine-tune Llama 3.1 8B-IT with offline Group Relative Policy Optimization (GRPO) on 5,160 trajectories from 29 biology training tasks, using cumulative bsf-AUC@15 as the per-trajectory reward (full setup in §B). The trained model improves bsf-AUC@15 on 14 of 21 held-out tasks (Wilcoxon $p = 0.019$), in-distribution on held-out biology and cross-domain on never-seen education (§B). GRPO with a bsf-AUC reward is therefore a tractable training target on Llama 3.1 8B-IT, and the trained behavior is not narrowly biology-specific. Whether the bsf-AUC gains reflect broader trajectory-level improvements is mixed (§B). Trainability on frontier or closed-weight models remains open since we did not pilot GRPO beyond Llama 3.1 8B-IT.

5 Limitations and assumptions

Mechanism. We do not isolate a single mechanism for why domain-aware LLMs underperform on biology bsf-AUC. *What is demonstrated.* Domain-aware proposals hug the literature prior at the proposal level (§A.22); on the categorical column of 6 literature-divergent biology tasks, domain-aware lags domain-agnostic on all 6 (§4.3); the §4.3 audit anchors this cross-subject pattern in published wet-lab outcomes on 6 documented cases, so it is not a pure oracle artifact. *What is weakened.* Oracle-proximity as a sufficient mechanism, since closer-to-literature proposals do not systematically receive higher oracle scores (§A.22, score-vs-density heterogeneous). Pure proposal-side numeric anchoring as a sufficient mechanism, since domain-aware explores *more* of the parameter space than domain-agnostic yet still scores worse (§A.14). The cross-subject sign flip (§4.3) and GP-UCB ordering (§A.12) further constrain candidates. *What is unresolved.* Whether domain-aware semantic reasoning is misaligned with the oracle’s featureless input space (representational mismatch); whether the bsf-AUC gap on the full 45-task biology panel is primarily categorical or distributed across numeric and categorical channels. Distinguishing these would require per-design oracle confidence stratification, ablations isolating numeric vs. categorical contributions, and held-out wet-lab validation against novel designs.

Scope and selection. Education is a 10-task validation set; the within-biology findings of §4.1, §4.2, and §4.3 do not depend on it. The current education set lacks adversarial tasks where domain-aware would be expected to harm; a sharper stress-test would require education tasks with literature claims the published data contradicts. Claude Opus 4.7’s 9-task safety-filter refusals (§A.8) bias its individual win rate upward; the panel-level estimate is robust to Opus exclusion (§A.6). The benchmark covers parameterized scalar-outcome optimization. Cross-experiment reasoning is out of scope, and the published wet-lab audit is biology-only. Our GRPO finding is a within-benchmark trainability result, not a deployment recommendation; bsf-AUC improvements did not consistently translate to broader trajectory-level gains (§B), so deployment-grade training would need multiple complementary metric guardrails.

6 Conclusion and discussion

LEAPBENCH tests the assumption that domain priors plus reasoning over iterative feedback deliver learning efficiency in iterative scientific design, on 55 tasks across eight LLMs. The central finding

is methodological. **Evaluation design choices materially change conclusions about LLM learning efficiency.** All three axes change conclusions. *What to measure.* Trajectory scoring and outcome scoring pick a different top-ranked model on 24 of 45 biology tasks, and outcome scoring overlooks efficiency gains that trajectory scoring captures. *What to compare against.* No LLM clears GP-UCB parity on biology bsf-AUC@30 under either prompt condition. The prompt manipulation moves performance within the GP-parity band, not across it. *What to ground against.* The domain-aware versus domain-agnostic prompt manipulation, on identical parameter spaces, reveals a cross-subject asymmetry. Domain-aware prompting helps on the 10 education tasks, where literature consensus aligns with the experimental optimum, but underperforms on the 45 biology tasks, where it may not. On 16 biology tasks where feedback is actionable, domain-aware prompting reduces the iter-30 published-best match rate by 10 percentage points. On 6 tasks where the literature-typical answer differs from the published best, domain-agnostic finds the published best more often on all 6.

Our methodological findings argue for evaluation practice, not specific deployment recommendations. Practitioners should evaluate their candidate LLM under bsf-AUC@k when iteration budgets are tight and under bsf-Outcome@k when many iterations are within budget, since metric choice can change the top-ranked model. A GP-UCB comparison is worthwhile for assessing whether an LLM optimizer adds value over a classical baseline. The published-wet-lab-grounded findings on 6 documented cases suggest practitioners may want to test domain-aware vs. domain-agnostic prompts on representative tasks before committing in autonomous-lab settings, particularly when literature-typical and experimental-peak values may diverge.

Release and broader impact. LEAPBENCH (dataset, oracles, trajectories, baselines, training and audit data) and the reproduction code are released on Harvard Dataverse and GitHub respectively. Full inventory in §A. LEAPBENCH is a benchmark, not a deployment tool: biology tasks span dual-use technologies, but oracles encode only already-published outcomes and exclude hazard-relevant endpoints (toxin potency, transmissibility), so marginal capability gain over the source literature is small. The GRPO recipe is dual-use too; we release oracles trained only on public outcome surfaces.

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A Benchmark details

A.1 Metric-disagreement effect sizes and number of improving steps (NIS, supporting)

Effect-size breakdown of the 26 outcome-vs-bsf-AUC@30 disagreements. Half are close swaps (rank-1 vs. rank-2, 50%). The rest are deeper (rank-3 in 31%, rank-4 or further in 19%; median rank-distance 1.5). On magnitude, 23% of disagreeing tasks involve a $\geq 5\%$ bsf-AUC gap between the bsf-AUC-winner and the bsf-Outcome-winner, 12% a $\geq 10\%$ gap, and 8% a $\geq 20\%$ gap. Per-task Kendall’s τ between the two rankings averages +0.66, and 24% of tasks have $\tau < 0.5$. No task shows a reversed ordering. Disagreement is distributed similarly across domains (biology 47%, education 50%; Fisher $p = 1.0$). A related horizon comparison: bsf-AUC@5 vs. bsf-AUC@30 disagree on 27% of tasks and bsf-AUC@5 vs. bsf-Outcome@5 on 36%. Process vs. endpoint already diverge at short horizons.

Iterations-saved on convergent disagreement subset. On the 45 biology tasks, 24 are bsf-AUC@30 vs. bsf-Outcome@30 disagreements. Of those, 14 have the bsf-AUC and bsf-Outcome winners ending within 1% of each other at iteration 30. On those 14, the bsf-AUC winner’s best-so-far curve reaches the near-tied peak at median iteration 7 while the bsf-Outcome winner’s reaches it at iteration 15. The bsf-AUC winner is faster on 9 of 14 tasks, tied on 2, slower on 3 (paired Wilcoxon $p = 0.014$).

Three-way metric agreement. For each of the 55 tasks (pooled biology + education panel for direct comparison across metrics), we identify the best model under three metrics: outcome (final score), bsf-AUC (learning efficiency), and NIS (improving steps). Across 8 models all three agree on the same best model on only 1 of 55 tasks (2%). Outcome and NIS disagree on 41 of 55 = 75% of tasks, similar to the outcome-vs-bsf-AUC rate of 26 of 55 = 47% on the same panel (vs. the biology-only 24 of 45 = 53% reported in §4.1). bsf-AUC and NIS are trajectory-shape summaries of the same best-so-far curve and track each other at the model level (8-model biology-only Spearman $r = 0.79$, $p = 0.02$), so we treat NIS as a robustness re-expression of bsf-AUC rather than an independent metric.

Leave-one-model-out robustness of the outcome-vs-bsf-AUC disagreement. Under the tie-aware strict count on the pooled 55-task panel (47%, 26 of 55; biology-only count is 24 of 45 = 53% in §4.1), removing any single model leaves the disagreement rate a substantial minority of tasks. Removing Llama (the weakest model) does not reduce disagreement, so the finding is not driven by a noisy low-capability model flipping rankings. Removing Opus yields 38% (21 of 55), confirming the result survives Opus’s selection-biased task coverage (§5). A permissive argmax counting variant gives a higher raw rate (62% full panel, 49–62% across leave-one-model-out exclusions) and yields the same qualitative conclusion; we use the stricter tie-aware count in the main text because it is the more conservative definition.

Robustness to run-level noise (between-trajectory). The leave-one-model-out check above addresses *between-model* sensitivity (38–49% range across single-model exclusions). To check *between-trajectory* (within-pair) sensitivity, we bootstrap-resample the 4 runs per (task, model, condition) triple ($n_{\text{boot}} = 5,000$) and recompute each task’s classification. We use modal ($\geq 50\%$) bootstrap classification as it matches the median-based point estimates used elsewhere. The count is 25 of 55 = 45.5%, essentially matching the strict-tie count. The bootstrap and LOM-O together

address both noise sources in the panel.

Per-model details (supporting). Table 1 reports per-model Δ NIS by R^2 stratum on biology for all eight models (Mann-Whitney U). This table is included for completeness. We do not read it as independent mechanism evidence (NIS tracks bsf-AUC at the model level, and the per-stratum Δ NIS is computed on the same trajectories that produce the domain-aware win-rate gap). Five of eight models show significant NIS compression on high- R^2 biology. GPT-5.4 Mini is the notable exception, showing NIS *expansion* under semantic framing, consistent with its comparatively weak prior.

Table 1. Per-model Δ NIS by R^2 stratum. Δ NIS = (domain-aware NIS) – (domain-agnostic NIS). Negative Δ means domain-aware prompting reduces improving steps.

Model	Δ NIS (Variable)	p	Δ NIS (Clean)	p
Claude Opus 4.7	−1.44	0.24	−1.22	2×10^{-3}
Gemini 3.1 Pro	+1.61	0.88	−0.97	2×10^{-7}
Gemini 3 Flash	−0.15	0.84	−1.07	2×10^{-21}
Claude Sonnet 4.6	+1.74	2×10^{-3}	−0.41	3×10^{-3}
GPT-5.4	+0.64	0.32	−0.01	0.04
Llama 3.1 8B-IT	+0.34	0.41	−0.47	1×10^{-8}
DeepSeek V3.2	−2.83	2×10^{-4}	+0.19	0.43
GPT-5.4 Mini	+1.88	4×10^{-6}	+0.91	3×10^{-9}

A.2 Biology task data sources

The 45 biology tasks aggregate experimental data from 375 unique published studies. Oracle models are derived predictors trained on these published data, released alongside the benchmark for reproducibility. The DOIs of source papers are listed per task below.

- `adcc_reporter_gene_assay` (4): <https://doi.org/10.1016/j.jviromet.2022.114564>, <https://doi.org/10.1038/s41467-024-44786-2>, <https://doi.org/10.1186/s12967-025-06107-z>, <https://doi.org/10.3389/fimmu.2022.972168>
- `adcp_target_phagocytosis` (11): <https://doi.org/10.1007/s00262-018-2179-z>, <https://doi.org/10.1007/s12672-025-02281-0>, <https://doi.org/10.1016/j.ebiom.2023.104663>, <https://doi.org/10.1038/sj.bjc.6605355>, <https://doi.org/10.1073/pnas.2517415122>, <https://doi.org/10.1080/19420862.2015.1022692>, <https://doi.org/10.1080/2162402X.2020.1859263>, <https://doi.org/10.1136/jitc-2019-000195>, <https://doi.org/10.1136/jitc-2024-010945>, <https://doi.org/10.1186/s13058-024-01785-x>, <https://doi.org/10.3389/fimmu.2024.1483617>
- `antibody_complement_activation_cdc` (10): <https://doi.org/10.1007/s12672-025-02281-0>, <https://doi.org/10.1038/s41598-023-42925-1>, <https://doi.org/10.1038/s41598-025-16461-z>, <https://doi.org/10.1080/19420862.2019.1690959>, <https://doi.org/10.1111/bjh.18438>, <https://doi.org/10.1111/cas.70325>, <https://doi.org/10.1126/sciadv.aea3737>, <https://doi.org/10.3389/fimmu.2022.889372>, <https://doi.org/10.3389/fimmu.2022.957874>, <https://doi.org/10.3390/antib9040063>
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- **sars_cov2_fab_binding_affinity(9)**: <https://doi.org/10.1016/j.jmb.2021.166956>, <https://doi.org/10.1038/s41467-021-25153-x>, <https://doi.org/10.1038/s41467-022-28528-w>, <https://doi.org/10.1038/s41467-024-45050-3>, <https://doi.org/10.1038/s41586-020-2571-7>, <https://doi.org/10.1038/s41586-021-03398-2>, <https://doi.org/10.1038/s41598-021-00684-x>, <https://doi.org/10.1038/s41598-022-19699-z>, <https://doi.org/10.1126/science.abn8863>
 - **sf9_protein_yield(11)**: <https://doi.org/10.1007/s11033-024-10136-0>, <https://doi.org/10.1023/B:CYTO.0000039894.27256.0f>, <https://doi.org/10.1038/s41598-024-65316-6>, <https://doi.org/10.1186/s12896-018-0434-1>, <https://doi.org/10.1186/s12985-021-01507-1>, <https://doi.org/10.1186/s13036-019-0206-z>, <https://doi.org/10.1186/s13036-025-00562-4>, <https://doi.org/10.1371/journal.pone.0308547>, <https://doi.org/10.3389/fbioe.2022.908509>, <https://doi.org/10.3389/fmicb.2023.1171500>, <https://doi.org/10.4014/jmb.1906.06022>
 - **streptomyces_avermitilis_avermectin_yield(13)**: <https://doi.org/10.1038/s41467-024-49987-3>, <https://doi.org/10.1038/srep25949>, <https://doi.org/10.1038/srep44567>, <https://doi.org/10.1111/1751-7915.13813>, <https://doi.org/10.1111/1751-7915.14319>, <https://doi.org/10.1111/1751-7915.14470>, <https://doi.org/10.1128/AAC.15.3.361>, <https://doi.org/10.1128/AAC.32.2.282>, <https://doi.org/10.1128/AEM.00473-21>, <https://doi.org/10.1186/s12934-015-0337-y>, <https://doi.org/10.4014/jmb.1503.03042>, <https://doi.org/10.5812/jjm.10366>, <https://doi.org/10.5812/jjm.8626>
 - **streptomyces_synthesis_antibiotic_yield(6)**: <https://doi.org/10.1038/s41598-024-76860-6>, <https://doi.org/10.1038/srep08740>, <https://doi.org/10.1128/aem.01133-23>, <https://doi.org/10.1186/s12934-024-02510-1>, <https://doi.org/10.1186/s12934-025-02843-5>, <https://doi.org/10.3389/fmicb.2020.00224>
 - **transaminase_amination_conversion(8)**: <https://doi.org/10.1002/cbic.202100123>, <https://doi.org/10.1002/cssc.202501501>, <https://doi.org/10.1016/j.jbiotec.2018.12.001>, <https://doi.org/10.1186/s12896-021-00713-7>, <https://doi.org/10.1186/s40643-021-00448-5>, <https://doi.org/10.3390/molecules25092140>, <https://doi.org/10.3390/molecules27217331>, <https://doi.org/10.3390/molecules28093895>
 - **vhh_nanobody_expression_ecoli(5)**: <https://doi.org/10.1134/S0006297924050134>, <https://doi.org/10.1186/s13104-018-3852-1>, <https://doi.org/10.1186/s13568-022-01422-w>, <https://doi.org/10.3389/fimmu.2022.1022418>, <https://doi.org/10.52601/bpr.2023.230026>

A.3 Prompt design

Domain-aware condition. The system prompt names the domain (e.g., “You are optimizing CRISPR HDR efficiency”), lists real parameter names with ranges and units, and specifies the optimization objective. Categorical parameters are listed by name (e.g., “cell_line: one of HEK293T, HEK293, RPE1, U2OS, MSCs”).

Domain-agnostic condition. The system prompt reads: “You are optimizing a black-box function. Propose parameter values to maximize/minimize the target.” Parameters are renamed X1, X2, ... (numerical) and C1, C2, ... (categorical, with options A, B, C, ...). Ranges are preserved but

units are removed. The oracle and scoring are identical in both conditions.

Example prompt pair (cho_antibody_titer). The full set of per-task prompt templates is in the code release. A representative example for one task follows.

Domain-aware system prompt:

You are an expert in CHO cell culture and monoclonal antibody production. You are helping design experiments to maximize monoclonal antibody titer (g/L).

When generating hypotheses, return ONLY a valid JSON array with these fields:

- hypothesis_name (string, 3-5 words)
- cho_cell_line (string: "CHO-K1", "CHO-S", or "CHO-DG44")
- cell_specific_perfusion_rate (number: 100-500 pL/cell/day)
- foxa1_overexpression (number: 0 = no, 1 = yes)
- feed_rate_strategy (string: "adaptive", "dynamic", or "continuous")
- rationale (string, 1 sentence explaining why this might increase titer)

Learn from the experimental feedback provided and generate improved hypotheses.

Domain-agnostic system prompt (same task):

You are optimizing a black-box function. Propose parameter values to maximize the target.

When generating hypotheses, return ONLY a valid JSON array with these fields:

- hypothesis_name (string, 3-5 words)
- C1 (string: "A", "B", or "C")
- X1 (number: 100-500)
- X2 (number: 0-1)
- C2 (string: "A", "B", or "C")
- rationale (string, 1 sentence)

Learn from the feedback provided and generate improved hypotheses.

The oracle and scoring function are identical. Only the prompt content and parameter names differ.

GRPO evaluation. All 21 held-out GRPO evaluation tasks use a uniform “maximize” objective to simplify the training signal, so the model learns a single optimization direction from feedback. The 3 tasks whose conventional objective is minimize (HER2 binding affinity, SARS-CoV-2 Fab binding, FcγR-enhanced binding) are evaluated as maximize tasks for this comparison. This does not reduce task validity since maximizing these metrics corresponds to a different but scientifically valid design goal (e.g., weaker binding for selectivity or safety). The main benchmark (§4) uses task-specific directions.

A.4 Robustness to an alternative BO baseline (HEBO)

GP-UCB is one classical baseline. We re-run the comparison against HEBO (Cowen-Rivers et al., 2022), the heteroscedastic-evolutionary BO that won the 2020 NeurIPS BBO challenge and is

a popular off-the-shelf default for mixed numeric+category spaces. Per-task HEBO best-so-far curves come from 4 runs at the same 30-iteration budget as the LLM and GP-UCB conditions (eval/baselines/run_hebo_4runs_parallel.py). Head-to-head on biology, HEBO wins 21 of 45 tasks at median bsf-AUC@30 against GP-UCB and loses 24, so the two baselines are comparable in strength on this panel rather than HEBO being uniformly stronger. Figure 6 mirrors Figure 3a, replacing GP-UCB with HEBO under both prompt conditions.

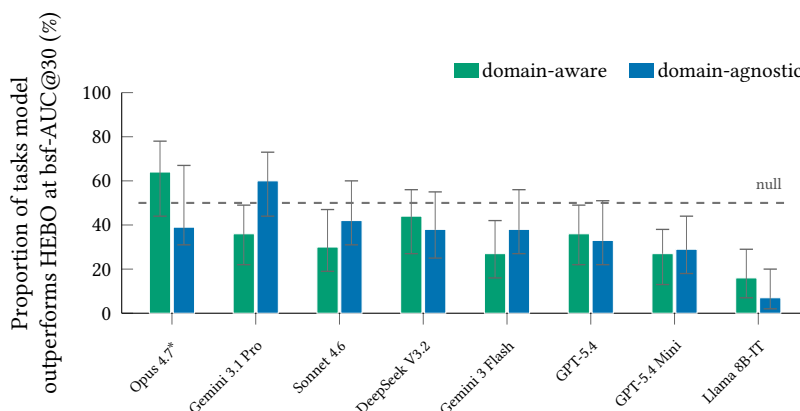


Figure 6. Per-model bsf-AUC@30 outperformance vs. HEBO on biology, both prompt conditions. Each model’s two bars give the fraction of biology tasks where its median bsf-AUC@30 outperforms HEBO’s, under domain-aware (teal) and domain-agnostic (cobalt). Dashed line marks the 50% null. Error bars are 2-level bootstrap 95% CIs (4-run-matched HEBO). Pass rates against HEBO move in mixed directions per model relative to the GP-UCB rates (Figure 3b, Figure 14). The pattern is consistent with HEBO and GP-UCB being task-specifically different rather than one uniformly stronger (head-to-head HEBO wins 21 / 45 biology tasks at median bsf-AUC@30; GP-UCB wins 24). The §4.2 “no model clears 50%” result holds against HEBO under both conditions, with the same single exception (Opus 4.7 under domain-aware, point estimate 64% [44, 78]; CI overlaps the null). *Opus’s coverage is 36 / 45 biology tasks (§A.8).

A.5 Statistical conventions and bootstrap run sensitivity

Conventions. Unless stated otherwise, each (model, task) point is the median of 4 per-run bsf-AUC values. Task-level confidence intervals come from a task-bootstrap that resamples the 55 tasks with replacement ($B = 1,000$). Per-model aggregates use a 2-level bootstrap (resample tasks, then resample runs within task). Reported bsf-AUC@ k is evaluated at iteration k of the best-so-far curve (not averaged over iterations $1..k$).

Run-level noise. To test the stability of the biology win rate under run-level sampling noise, we resample 4 runs with replacement for every (model, task, condition) triple on biology and recompute the domain-aware win rate. Over 1,000 bootstrap iterations, the resampled point estimate stays near 42.4% in every resample (337 pairs). Run-level noise does not materially move the point estimate.

A.6 Leave-one-model-out robustness of the biology win rate

To check that the biology win rate is not driven by any single model, we recompute the domain-aware win rate eight times, excluding one model per run. Every exclusion leaves the point estimate in the 40.7–43.8% range:

The point estimate is robust across exclusions, staying in a narrow 40.7–43.8% band.

Table 2. Biology domain-aware win rate under leave-one-model-out exclusion. Domain-aware win rate on biology recomputed eight times, excluding one model each time. Task-clustered 95% CIs shown.

Excluded model	domain-aware win rate	Task-clustered 95% CI	Clustered p
Claude Opus 4.7	40.7%	[31.4, 50.8]	0.070
Claude Sonnet 4.6	42.9%	[33.7, 52.5]	0.147
GPT-5.4	41.4%	[31.8, 51.8]	0.104
Gemini 3.1 Pro	43.8%	[34.1, 54.0]	0.236
DeepSeek V3.2	42.6%	[33.2, 52.5]	0.143
Gemini 3 Flash	43.2%	[33.8, 53.0]	0.172
GPT-5.4 Mini	43.2%	[33.8, 53.1]	0.175
Llama 3.1 8B-IT	41.8%	[31.8, 52.4]	0.130
<i>Full panel</i>	42.4%	[33.1, 52.4]	0.135

A.7 Model inference

All models are queried via OpenRouter (anthropic/claude-sonnet-4-6, anthropic/claude-opus-4, google/gemini-3.1-pro-preview, google/gemini-3-flash-preview, openai/gpt-5.4, openai/gpt-5.4-mini, deepseek/deepseek-v3.2, meta-llama/llama-3.1-8b-instruct) at provider-default sampling (temperature = 1.0, no top- p or top- k override) with max_tokens = 16,384. Each model proposes one candidate design per iteration as a JSON object. Parse failures are retried up to twice with a clarifying instruction to return valid JSON only. Sampling is unchanged on retry. Persistent failures receive the worst possible score for that task. Across the full canonical-source corpus reported in this paper (13,570 trajectories totaling 392,099 iterations), 0 iterations resulted in persistent parse failures after retry; the worst-score fallback was never triggered, so parsing reliability is not a hidden confound on bsf-AUC. We run 4 independent runs per model per task per condition (domain-aware and domain-agnostic) over 30 iterations each. Safety-filter refusals on a subset of domain-aware biology tasks (§ A.8) reduce coverage to 412 complete (model, task) pairs.

Compute footprint. The full canonical evaluation matrix is 13,570 trajectories and 392,099 LLM iterations (8 models $\times$ 55 tasks $\times$ 2 conditions $\times$ 4 runs $\times$ 30 iterations, minus refusals). GP-UCB baselines comprise 200 runs $\times$ 55 tasks (200-run baseline) plus 4 runs $\times$ 55 tasks (4-run-matched, used in Figures 2(b), 3(b), 6); both run in <5 minutes of CPU. Per-call OpenRouter cost is included in the raw run JSON, so reproducers can recompute aggregate spend from the released data.

A.8 Provider refusals

Claude Sonnet 4.6 and Claude Opus 4.7 exhibit safety-classifier refusals on domain-aware prompts containing pathogen-target, antibody-binding, antimicrobial, or surfactant-chemistry terminology. These return empty responses or refuse to generate parameter sets even under benign optimization framing. Claude Sonnet is affected on 2 tasks (microemulsions_particle_size, sars_cov2_fab_binding_affinity). Claude Opus on 9 tasks (FcγR binding, HER2 binding, SARS-CoV-2 Fab binding, *E. coli* MIC, *S. aureus* biofilm, antibody complement activation and cytotoxicity, microemulsion zeta-potential drug-delivery). Domain-agnostic runs of the same tasks (parameters renamed X1, C1) completed normally. We exclude these (model, task) combinations from the relevant comparisons, yielding 53 task pairs for Claude Sonnet, 46 for Claude

Opus, and 337 complete (model, task) biology pairs across all eight models. No other (model, task) combinations were affected. See §5 for the selection-bias implication.

A.9 Disagreement and rank are unstable across horizons

The bsf-AUC@ k vs. bsf-Outcome@ k disagreement is stable across horizons, and per-model rank under bsf-AUC@ k outperformance vs. GP-UCB is not. Figure 7 shows the tie-aware-strict disagreement rate on biology at $k \in \{5, 10, 15, 20, 25, 30\}$ (same definition as the 53% result at $k = 30$). Disagreement rises with horizon, from 36% at $k = 5$ to 53% at $k = 30$, never falling below 36%. Figure 8 ranks each model by the fraction of biology tasks where its median bsf-AUC@ k outperforms GP-UCB. GPT-5.4 leads at $k = 5$ and falls to rank 4 by $k = 30$. DeepSeek V3.2 climbs from rank 5 to rank 2. Gemini 3 Flash drifts from rank 3 to rank 6. The other five models stay within one rank of where they start. A practitioner choosing the best model after a 5-iteration budget would deploy a different system than one choosing after 30.

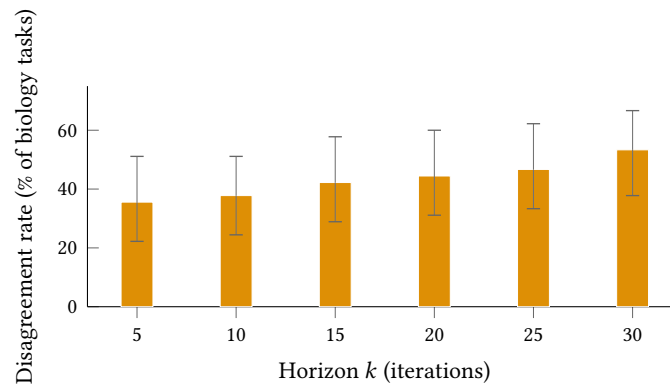


Figure 7. Disagreement rate between bsf-AUC@ k and bsf-Outcome@ k across horizons (biology, 45 tasks). Bars show the fraction of biology tasks where the argmax-of-median bsf-AUC@ k winner is not in the tied-best set under bsf-Outcome@ k (canonical tie-aware-strict rule). Error bars are task-clustered bootstrap 95% CIs ($B = 2000$).

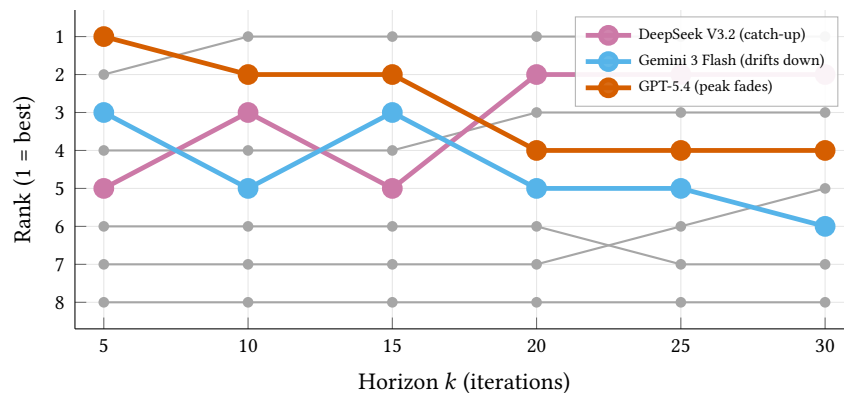


Figure 8. Per-model rank on biology bsf-AUC@ k vs. GP-UCB, across horizons. Each model is ranked by the fraction of 45 biology tasks where its median bsf-AUC@ k outperforms GP-UCB. Lines connect the same model across $k \in \{5, 10, 15, 20, 25, 30\}$. The three highlighted models have non-trivial rank movement.

A.10 Education companion: metric-task flip across horizons

Companion to §A.9 on the 10-task education panel. The tie-aware-strict bsf-AUC@ k vs. bsf-Outcome@ k disagreement rate is in the same range as biology, between 20% and 50% across $k \in \{5, \dots, 30\}$ (Figure 9). The smaller panel ($n = 10$) gives wider task-clustered bootstrap CIs, but the headline finding from §4.1, that metric choice flips per-task winners on a non-trivial fraction of tasks, holds in the second domain.

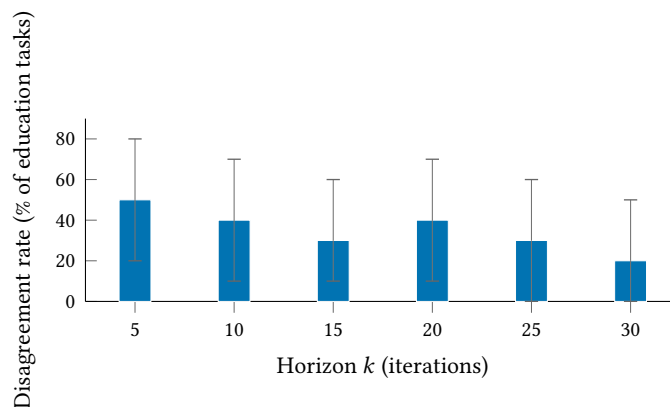


Figure 9. Disagreement rate between bsf-AUC@ k and bsf-Outcome@ k across horizons (education, 10 tasks). Companion to Figure 7. Bars use the same tie-aware-strict rule. Error bars are task-clustered bootstrap 95% CIs ($B = 2000$). The smaller panel widens CIs, but flip rates remain non-trivial at every horizon.

A.11 Best-model confusion matrix: bsf-AUC@30 vs. bsf-Outcome@30

Figure 10 shows the per-model breakdown of the 24 biology disagreement tasks. Most concentrate on tasks where Claude Opus 4.7 or Gemini 3.1 Pro wins bsf-Outcome@30 while a faster-learning model (Sonnet, DeepSeek, or Pro) wins bsf-AUC@30.

A.12 GP-UCB does not saturate biology tasks

The biology finding is not a saturation artifact. Under a saturation account, if GP-UCB saturates easy tasks, both LLM conditions would pile up at the top and the 42% domain-aware win rate would reflect noise, not literature-prior stickiness. The data rule this out. On biology (337 pairs) the median GP-normalized bsf-AUC is -0.06 under domain-aware and -0.02 under domain-agnostic. Both conditions fall *below* GP-UCB, and domain-aware further so, consistent with prior-driven commitment pushing models below what the signal would otherwise support. Only 33% of (task, model) pairs under domain-aware and 39% under domain-agnostic reach positive GP-normalized bsf-AUC. No ceiling saturation.

A.13 Recency check: GPT-5.5

GPT-5.5 was released April 23, 2026, two weeks before submission and after our main evaluation cycle had completed. Rather than rebuild the full canonical with a 9th model under deadline, we ran a targeted recency + stress test on 13 tasks chosen for high leverage on specific claims: the 6 literature-divergent biology tasks that drive the §4.3 audit, polymeric_micelles_drug_loading

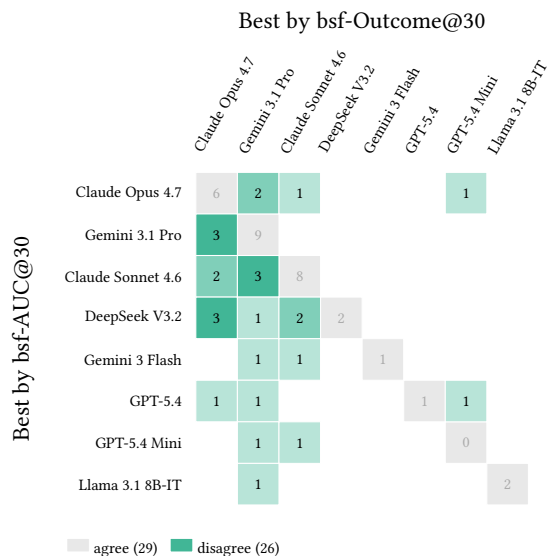


Figure 10. Best-model confusion matrix over 55 tasks. Diagonal (gray) = agreement (29 tasks). Off-diagonal (green, shade $\propto$ count) = 26 tasks where bsf-AUC@30 and bsf-Outcome@30 pick different best models.

and `ottmar_perceptual_cues` as biology-non-divergent and education representatives, and 5 biology tasks where the closest-running LLM came within 3% of GP-UCB AUC, chosen as adversarial cases for the §4.2 majority-trail claim. All matched conditions: 4 runs $\times$ 30 iterations $\times$ paired domain-aware/domain-agnostic, identical prompt template.

Sample efficiency vs. GPT-5.4. On the 8 common tasks, GPT-5.5 wins on 4 of 8 in raw bsf-AUC under both conditions (Wilcoxon paired $p = 1.00$ domain-aware, $p = 0.74$ domain-agnostic). Median GP-normalized bsf-AUC is essentially flat between the two versions, well within run-to-run noise.

Literature-divergent reversal (§4.3). Adding GPT-5.5 to the panel yields a 9-model evaluation on the 6 divergent tasks. The reversal still holds. Averaged across models per task, domain-aware selects the published best on 23.0% vs. 36.0% for domain-agnostic (paired Wilcoxon across tasks, $p = 0.031$, 6 of 6 tasks reversed). GPT-5.5 alone shows the same anchoring pattern qualitatively (domain-aware 16.7% vs. domain-agnostic 37.5%, mean $\Delta = -20.8\%$), but with domain-aware = domain-agnostic = 0% ties on 4 of 6 tasks no single-model paired test reaches significance.

Modal-rank gap (§A.28). The OLS test on miss-conditional rank with (model, task)-clustered standard errors remains significant in the 9-model panel. Coefficient -0.56 , 95% CI $[-1.07, -0.06]$, $p = 0.030$ ($n = 265$ misses across 48 (model, task) clusters), essentially unchanged from the 8-model result (-0.64 , $p = 0.029$).

Both methodological and published-wet-lab-anchored audit findings are robust to including the most recent frontier model.

GP-UCB stress test (§4.2). On the 5 closest-margin biology tasks (`microemulsions_particle_size`, `mammalian_crispr_hdr_eff`, `qpcr_efficiency`, `hydrogels_release_kinetics`, `antibody_complement_dependent_cytotoxicity_cdc`), GPT-5.5’s bsf-AUC@30 (best of domain-aware/domain-agnostic) clears GP-UCB on 1 of 5 (`antibody_cdc`, the largest

pre-existing margin). Combined with the recency-check coverage, GPT-5.5 outperforms GP-UCB on 4 of 12 tested biology tasks (33%). Extrapolating to the full 45-task biology panel at this rate gives ≈ 15 expected wins, well below the 22-of-45 majority threshold needed to break the §4.2 claim. The 13-task subset is not a random sample. It was chosen to threaten specific claims, so these numbers are adversarial stress tests, not full-panel point estimates. The 1-of-5 stress-test result and 4-of-12 combined result are unchanged when re-evaluated against the 4-run-matched GP baseline used in main figures.

Cross-subject domain-aware-domain-agnostic gap (§4.3). Adding GPT-5.5 on its 13-task subset moves the cross-subject win-rate gap from 31.3 percentage points to 32.3 (biology: 42.4% $\rightarrow$ 41.8%; education: 73.7% $\rightarrow$ 74.0%). GPT-5.5 alone shows the same qualitative pattern more sharply (3 of 12 = 25% biology domain-aware wins; 1 of 1 = 100% education).

Metric-disagreement count (§4.1). Adding GPT-5.5 on its 13-task subset shifts the strict-tie disagreement count from 27 of 55 to 28 of 55 (one task flips from agree to disagree). Other panel-level numbers are unchanged on the 42 tasks where GPT-5.5 has no data.

A.14 Parameter-space diversity

If domain-aware biology models simply explored less of the parameter space, that alone could produce the reversal. Parameter-space diversity (mean pairwise L2 distance between per-iteration designs in $[0, 1]$ -normalized space) rules this out. On biology, in a paired per-(task, model) comparison, domain-aware diversity exceeds domain-agnostic on 204 of 267 high- R^2 (task, model) combinations (76%) (Wilcoxon signed-rank $p = 7 \times 10^{-21}$; means 0.61 vs. 0.40). The same pattern holds on low- R^2 biology (15 of 15, 100%; $p = 6 \times 10^{-5}$) and across all biology (225 of 290, 78%; $p = 4 \times 10^{-24}$). Yet domain-aware scores are worse on high- R^2 biology. Despite *greater* exploration, domain-aware models still underperform, suggesting *misdirected* exploration. Domain-aware models explore widely within the space of “known-good” designs rather than toward what actually works for the specific task (Figure 11).

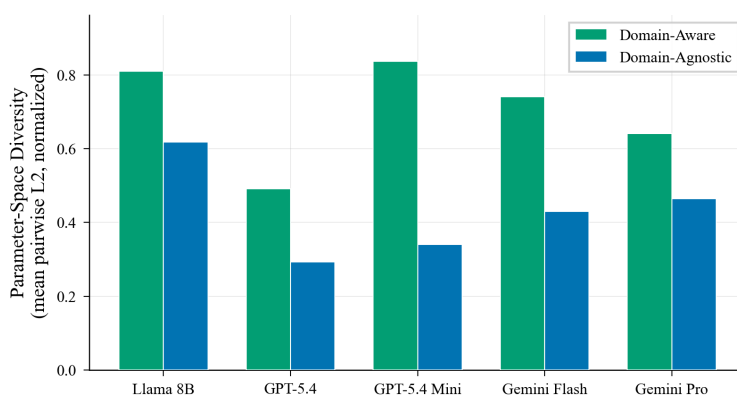


Figure 11. Exploration is not the missing ingredient. Domain-aware diversity is higher on $\sim 80\%$ of (task, model) combinations, yet domain-agnostic runs achieve better scores on high- R^2 tasks.

A.15 Robustness of Figure 4: alignment, match definition, and inferential tests

This appendix gives the full robustness battery for the oracle-aligned match-rate finding in §4.3.

Oracle-alignment criterion. For each biology task with categorical features, we load the surrogate oracle pickle (sklearn pipeline) and score every row in the published `dataset.csv`. The *published-best* row is the row with the highest measured target value (or lowest, for minimize tasks). We define the alignment ratio as $(s_{\text{wlb}} - s_{\text{min}})/(s_{\text{max}} - s_{\text{min}})$ for maximize tasks, with the analogous $(s_{\text{max}} - s_{\text{wlb}})/(s_{\text{max}} - s_{\text{min}})$ for minimize tasks, where s_{wlb} is the oracle’s predicted score for the published-best row, and $s_{\text{min}}, s_{\text{max}}$ are the oracle’s predictions over the dataset rows. A ratio of 1.0 means the oracle’s argmax row coincides with the published-best row. The primary threshold is ≥ 0.95 . Sensitivity to the threshold is reported below.

Key categorical per task. For each task, we compute the oracle-score spread (max minus min mean prediction across observed values) for each categorical column. The *key categorical* is the column with the largest spread. We then check whether the published-best row has a non-NaN value in that column; if not, we descend the spread-ranked list until we find a column with a non-NaN published-best value. The single-key-cat-per-task definition gives a tighter, lower-variance match rate than the any-cat-match definition (see §4.3, where the any-cat-match averaging is appropriate because the targeted finding is the literature-divergent subset, not key-cat alignment).

Threshold sensitivity. At the primary threshold ≥ 0.95 , $n = 16$ biology tasks pass; at ≥ 0.90 , $n = 18$; at ≥ 0.99 , $n = 13$. The mixed-effects iter-30 effect is stable: -10.4 percentage points ($p = 0.0011$ one-sided) at ≥ 0.90 , -10.4 percentage points ($p = 0.0037$) at ≥ 0.95 , -11.2 percentage points ($p = 0.0045$) at ≥ 0.99 . The climb effect is similarly stable: -15.3 percentage points ($p < 0.001$) at ≥ 0.90 , -11.3 percentage points ($p = 0.0039$) at ≥ 0.95 , -7.5 percentage points ($p = 0.058$, marginal) at ≥ 0.99 where n shrinks the most.

Mixed-effects model. Linear probability model on the per-(task, model, condition) cell mean iter-30 (or iter-30 minus iter-1 climb) match rate, fit with crossed random intercepts for task and model: `outcome ~ condition + (1 | task) + (1 | model)`. We use `statsmodels.MixedLM` with the variance-component formulation that supports crossed random effects. Coefficient on the domain-aware indicator is -0.104 (SE 0.039 , two-sided $p = 0.0074$, one-sided $p = 0.0037$) for iter-30 and -0.113 (SE 0.042 , two-sided $p = 0.0078$, one-sided $p = 0.0039$) for the climb at the primary threshold.

Permutation test. Within each (task, model) pair, the DA and DG cell means are exchangeable under the null. We swap the DA/DG label per cell with a Rademacher sign and recompute the mean diff over all 111 cells. Over $B = 5,000$ permutations: iter-30 one-sided $p = 0.0044$, climb one-sided $p = 0.014$.

Two-way cluster bootstrap. Resample tasks with replacement ($B = 2,000$); recompute the per-cell paired diff and its mean. The 95% CI for the iter-30 mean diff is $[-0.225, +0.029]$ with point estimate -0.105 , and the climb mean diff is $[-0.212, +0.015]$ with point estimate -0.102 . The bootstrap distributions place the iter-30 diff below zero in 93% of resamples and the climb diff below zero in 96%.

Per-task and per-model paired tests (descriptive). Paired Wilcoxon over the 16 task-level mean diffs (averaging across models) gives one-sided $p = 0.064$ for iter-30 and $p = 0.044$ for the climb, marginal at $\alpha = 0.05$ with $n = 16$. The per-model sign test on diffs (averaging across tasks) is 5 of 8 negative ($p = 0.36$) for iter-30 and 5 of 8 ($p = 0.36$) for the climb at the primary threshold; the

cross-model summary is therefore weaker than the cell-level tests, consistent with the bootstrap finding that any single model’s task average has wide CI overlap. The cell-level mixed-effects model is the primary test because it pools the (task, model) cells while accounting for crossed dependence; the per-model summary appears only as a descriptive readout.

Caveats. (i) The oracle-alignment filter uses the dataset rows as the parameter-space sample, not a continuous grid; for tasks with sparsely sampled categoricals, the alignment ratio undercounts coverage. (ii) Selecting the key categorical by oracle-score spread privileges the column where the oracle has the most signal; this is the column where feedback can in principle steer the model, which is the right inferential frame for the figure but not necessarily the categorical that researchers would pick on a priori grounds. (iii) The mixed-effects LPM gives an interpretable additive-percentage-point coefficient but ignores the binomial nature of the outcome; a binomial GLMM gives qualitatively similar coefficients (not shown). (iv) Claude Opus 4.7’s domain-aware coverage is reduced to 9 fewer tasks by the safety-filter refusals (§A.8). Including Opus does not change the main result (mixed-effects $p = 0.004$ with Opus included; $p = 0.003$ excluding Opus). Opus is the per-model exception, but the cross-model finding does not depend on it.

A.16 Education-mechanism identification: the symmetric counterpart to §4.3

The §4.3 audit asks whether domain-aware LLMs anchor to literature-typical values on the biology subset where literature-typical and published-best diverge. Here we apply the same machinery to education and ask the symmetric question: do domain-aware LLMs identify the RCT-confirmed correct mechanism more often than domain-agnostic, on tasks where literature consensus and best-result coincide?

Setup. 9 of 10 education tasks have at least one treatment categorical with an RCT-confirmed correct mechanism (oulad_vle_engagement excluded as observational with no treatment categorical). For each (task, model, condition, run) trajectory, we compute the trajectory’s modal categorical value and compare against the correct mechanism (the categorical value of the row with the best target in the per-task literature dataset; cross-checked against published RCT findings).

Per-task match-rates (modal categorical = correct mechanism, averaged across 4 runs × 8 models):

Task	domain-aware	domain-agnostic	Δ (domain-aware – domain-agnostic)
star_class_size	0.969	0.533	+0.435
pretesting_impossible	0.739	0.500	+0.239
ottmar_perceptual_cues	0.750	0.516	+0.234
assistments_experiments	1.000	0.893	+0.107
assistments_mastery_threshold	0.267	0.259	+0.007
interleaved_retrieval_practice	1.000	1.000	0.000
fractions_sequencing_nc	0.516	0.562	−0.046
interim_testing_categories	0.938	1.000	−0.062
retrieval_practice_wm	0.688	0.797	−0.109

domain-aware-better: 5 / domain-agnostic-better: 3 / tie: 1. Mean $\Delta = +0.089$. Per-task paired Wilcoxon two-sided $p = 0.31$ (one-sided $p = 0.16$); the per-task test is underpowered at $n = 9$.

Iteration-level pooled rate. Pooled across all 19,825 (task, model, condition, run, iter) education iterations, domain-aware proposes the correct mechanism on 66.4% of iterations vs. 60.9% for domain-agnostic ($\Delta = +5.5$ percentage points).

Per-iteration trajectory. The domain-aware edge concentrates in early iterations and decays as domain-agnostic learns the mechanism from oracle feedback:

Iteration window	domain-aware rate	domain-agnostic rate	Δ
Iter 0 (pre-feedback)	82.2%	50.6%	+31.6
Iter 0–9	66.2%	52.8%	+13.5
Iter 10–19	65.6%	64.7%	+0.9
Iter 20–29	66.1%	66.5%	−0.4

domain-aware enters with the correct mechanism (iter 0: 82% vs 51%), pulls away through iter ~ 6 , then domain-agnostic catches up by iter ~ 22 . The iter-0 difference is the cleanest piece of evidence: at iteration 0 there is no oracle feedback yet, so the domain-aware-domain-agnostic gap there is purely prompt-driven.

Caveat (oracle-shaping after iter 0). Iter-0 is oracle-independent at the LLM behavior level. Subsequent iterations are oracle-shaped: the LLM’s choices respond to oracle feedback, and the education oracle was trained on the same RCT data that defines the correct mechanism. So later-iteration match-rates conflate two stories. “domain-aware’s prior matches the RCT-confirmed mechanism” and “domain-aware’s prior matches what the oracle scores high” produce identical observed behavior here, because oracle and RCT agree. A cleaner test would use mechanisms confirmed by RCTs not in the oracle’s training set. The iter-0 gap is the cleanest oracle-independent signal.

Mechanism-level synthesis (across §4.3 and this audit). LLMs apply literature priors in both domains. On education, the prior matches the published-best (literature consensus is built from RCT meta-analyses) and domain-aware wins. On biology’s literature-divergent subset, the prior diverges from the published-best and domain-aware loses 6 of 6. The cross-subject reversal (§4.3) is therefore not two different mechanisms, but the same prior-application mechanism evaluated against two different alignment regimes.

A.17 Threshold sensitivity of the audit’s main result

The audit selects 6 literature-divergent biology tasks using two gap criteria connected by an OR: **(R)** top-1 vs. runner-up gap $\geq 10\%$ of target range, **(S)** top-1 vs. best literature-typical-row gap $\geq 0.5\sigma$. Sweeping the threshold confirms the reversal direction is not contingent on this single cut.

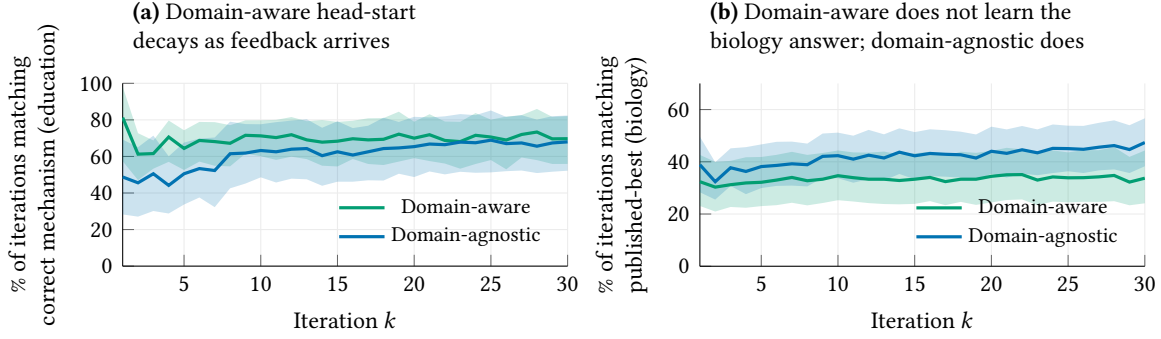


Figure 12. Cross-subject view of the prior-application mechanism. (a) **Education head-start decays.** Per-iteration % of education iterations where the trajectory’s proposal matches the RCT-confirmed correct mechanism (9 tasks). Domain-aware enters above and the gap closes by iter ~22 as oracle feedback teaches domain-agnostic the same mechanism. (b) **Biology domain-awareness does not improve over iterations.** Same metric on 34 biology tasks where the published best is identifiable. Domain-agnostic stays above throughout. The two panels visualize the cross-subject sign flip described in the synthesis paragraph above and in §4.3. Shaded bands are task-clustered bootstrap 95% CIs.

Cut	<i>N</i> pairs	Reversal (domain-agnostic > domain-aware)	Wilcoxon <i>p</i> (two-sided)	domain-aware / domain-agnostic mean
<i>Range only (gap to top-2 / target range):</i>				
≥ 0.05	7	6/7	0.297	0.31 / 0.33
≥ 0.10	5	5/5	0.062	0.24 / 0.41
≥ 0.15	3	3/3	0.250	0.31 / 0.47
≥ 0.20	3	3/3	0.250	0.31 / 0.47
≥ 0.25	3	3/3	0.250	0.31 / 0.47
<i>Std only (gap to best literature-typical-row / target σ):</i>				
≥ 0.25 σ	7	6/7	0.297	0.31 / 0.33
≥ 0.50 σ	4	4/4	0.125	0.23 / 0.38
≥ 0.75 σ	4	4/4	0.125	0.23 / 0.38
≥ 1.00 σ	4	4/4	0.125	0.23 / 0.38
<i>Either (paired range OR std):</i>				
R ≥ 0.05 OR S ≥ 0.25	7	6/7	0.297	0.31 / 0.33
R ≥ 0.10 OR S ≥ 0.50 (published)	6	6/6	0.031	0.21 / 0.36
R ≥ 0.15 OR S ≥ 0.75	4	4/4	0.125	0.23 / 0.38
R ≥ 0.20 OR S ≥ 1.00	4	4/4	0.125	0.23 / 0.38

Direction. Across all non-trivial cuts ($N \geq 4$), every qualifying (task, categorical) pair shows domain-agnostic > domain-aware on best-result match. **Power.** Two-sided $p \leq 0.10$ holds at $N \in \{5, 6\}$; stricter cuts shrink N to 3–4 and lose power even though direction is unanimous. **Permissive cuts.** The most permissive cut ($R \geq 0.05$ or $S \geq 0.25\sigma$) admits a 7th pair, `qpcr_efficiency / detection_chemistry`, which reverses sharply (domain-aware = 0.67, domain-agnostic = 0.36). The 6/6 unanimity at the published cut is therefore robust to tightening the gap thresholds but not to loosening them below the published bound. We do not read this as undermining the finding. The published cut is a principled selection (defined a priori as “literature-typical and best-result diverge by a meaningful margin”), the GPT-5.5 robustness check (§A.13) preserves the

6/6 unanimity under a 9-model panel, and direction-unanimity at $N = 6$ across our defined cut is what we report. A reader could equivalently describe the published cut as “the threshold above which the one task that breaks the pattern is excluded,” and that framing is consistent with the data.

The full sweep with all metric variants and the script that produced this table are at `paper_scripts/threshold_sensitivity`. Results are saved to `paper/threshold_sensitivity.json`.

A.18 Prompt-template robustness on the literature-divergent reversal

The 6/6 reversal in §4.3 uses one prompt template. We re-ran the audit on the 6 literature-divergent tasks under three perturbations: **V1_reorder** shuffles categorical-option order, **V2_terse** strips system-prompt verbosity, and **V3_altframe** replaces the opening framing. Sweep: 6 tasks $\times$ 4 models (Opus 4.7, Gemini 3.1 Pro, GPT-5.4, Llama 3.1 8B) $\times$ 2 conditions $\times$ 3 variants $\times$ 1 run = 144 trajectories, scored by the same all-30-iteration published-best match rate as the main audit; V0 (4 runs $\times$ 8 models, 6/6 reversal) is the comparator. Per-(variant, task) means and gaps are in `paper/fig_prompt_sensitivity.dat`.

Direction-preserving with similar magnitude to V0. Domain-agnostic $>$ domain-aware on 13 of 18 (variant, task) cells (one-sided binomial $p = 0.048$). Mean domain-agnostic – domain-aware gap: +10.6 percentage points, comparable to V0’s iter-30 gap of +15.4 percentage points on the same 6 tasks. Per variant: V1_reorder 4/6, V2_terse 4/6, V3_altframe 5/6.

Per-model. Direction holds majority for the three frontier models: Opus 4.7 13/18 cells, Gemini 3.1 Pro 12/17, GPT-5.4 9/18. Llama 3.1 8B is the prompt-template-sensitive outlier at 4/13, consistent with its smaller-model behavior elsewhere in the audit.

Per-task. Direction-preserved on 4 of 6 tasks (microalgae +52.3, vhh +11.7, transaminase +8.7, chl_yn ≈ 0); 2 of 6 reverse on average across variants (cho_antibody -5.0 , hek293_prime_editing -4.2). The V0 6/6 unanimity does not strictly survive prompt perturbation, but the headline direction (domain-agnostic $>$ domain-aware) and effect magnitude do. Per-cell rather than per-task analysis is reported because, with 1 run and 6 tasks, task-level tests have very low power.

A.19 Oracle reward by literature-modal rank

For each of the 6 literature-divergent biology tasks, we ranked the divergent categorical’s unique values by literature-frequency rank (rank 1 = literature-typical) and queried the oracle for each value’s reward holding all other parameters at the dataset’s per-column median (numerics) or modal value (other categoricals). This isolates each value’s oracle-implied effect from co-variation in other parameters.

On 4 of the 6 tasks (hek293, vhh_nanobody, chl_yn, cho_antibody) the oracle’s reward for the published best exceeds its reward for every literature value more modal than the published best. The remaining 2 fail for benign reasons. Microalgae’s oracle gap between photobioreactor and raceway pond is 0.003 in target range, effectively a tie; transaminase’s random-forest oracle is degenerate over cosolvent_type, scoring 91.72 for every value. On the 4 informative tasks, the oracle’s reward for the published best is at or near the argmax across all values and exceeds every more-modal literature alternative, so the feedback channel can in principle distinguish the published best from the literature-typical answer.

Table 3. Oracle reward by literature-modal rank on the 6 literature-divergent tasks. “Top-modal” is the most-frequent literature value (rank 1); “best” is the published-best value with its literature-modal rank in parentheses. Reward units are task-specific (target column).

Task	Top-modal value	Reward	Published best (modal rank)	Reward
hek293_prime_editing_indel_freq	transfection	22.98	Baculovirus (rank 4)	23.54
microalgae_biomass_multi_strain	photobioreactor	0.749	open raceway pond (rank 2)	0.746
vhh_nanobody_expression_ecoli	pET-22b	22.66	pCold-I (rank 4)	22.70
chl_yn_cell_line_optimization	dynamic (tie)	2.70	adaptive (rank 1, tied)	3.40
cho_antibody_titer	CHO-S	3.00	CHO-K1 (rank 2)	3.02
transaminase_amination_conversion	toluene	91.72	choline chloride mix (rank 6)	91.72

A.20 Literature-stickiness persists across model sizes despite training-data access

The literature-divergent tasks (§4.3) require finding designs better than what the literature suggests is typical. We test whether the literature-divergent failure could be a knowledge gap rather than a prior-overrides-feedback issue by checking whether the source papers predate model training. For each of the 6 tasks we identified the publication date of the source paper for the published-best row: microalgae_biomass (2018-04), chl_yn_cell_line (2020-10), cho_antibody_titer (2021-05), transaminase_amination (2023-05), vhh_nanobody_expression (2024-05), and hek293_prime_editing (2026-02). Five of the six predate every model’s release date by more than a year, so for those tasks the published-best paper should be in training data. Only HEK293 postdates a few model releases.

Conservative training-cutoff test. Restricting to (model, task) pairs where the source paper precedes the model’s release date by at least 12 months, a conservative proxy for being inside the training cutoff, gives 277 iterations across 39 (model, task) pairs (8 models, average ≈ 5 tasks each). On this subset, domain-aware iterations select the literature-typical value at 24.0% vs. 15.5% for domain-agnostic (Fisher one-sided $p = 0.052$, odds ratio 1.71).

Holds across all model size tiers. The domain-aware-greater-than-domain-agnostic direction is preserved in every tier: frontier models (Opus 4.7, Sonnet 4.6, GPT-5.4, Gemini 3.1 Pro): 14.7% vs. 11.4% (+3.3 percentage points); mid-tier (GPT-5.4 Mini, Gemini 3 Flash, DeepSeek V3.2): 35.6% vs. 18.3% (+17.2 percentage points, tier-only Fisher $p = 0.039$); small (Llama 3.1 8B-IT): 44.4% vs. 33.3% (+11.1 percentage points). The pattern is therefore not concentrated in smaller or older models, supporting the search-off design choice (§5). Enabling retrieval at inference would let the model look up the better paper, but on these tasks the better paper is already in training and the model still defaults to literature-typical.

Robustness to oracle alignment. We compared the oracle’s score for the published-best row to the oracle’s max score over the published dataset. On 4 of 6 literature-divergent tasks the published-best row is at or near the oracle max (vhh 1.00, hek293 1.00, transaminase 0.93, cho 0.80). On the other 2 it isn’t (microalgae 0.60, chl_yn 0.38), with the oracle assigning higher scores to other parameter combinations. The 6/6 domain-aware-vs-domain-agnostic reversal holds in both subsets (4/4 + 2/2).

A.21 Forced-search robustness check

§A.20 tests retrieval the static way (whether the source paper is in training data). This check tests retrieval the dynamic way: with web search enabled at inference, can the LLM look up the published best? We re-ran Claude Opus 4.7 on the 6 literature-divergent tasks with web search forced via OpenRouter’s `tool_choice = 'any'` setting and the `web_search_20250305` tool, which guarantees a search call at every iteration. We logged at minimum 240 web-search calls per task across the 4 runs $\times$ 30 iterations.

Table 4. Per-iteration published-best match rate on the 6 literature-divergent tasks under Claude Opus 4.7 only, search-OFF (canonical) vs. search-ON (forced). % top1 = fraction of iterations across 4 runs $\times$ 30 iterations where the proposed value matches the published best. “domain-agnostic minus domain-aware” is the domain-agnostic minus domain-aware difference.

Task	D-aware off	D-agnostic off	D-aware on	D-agnostic on	(agn – aware) off	(agn – aware) on
hek293_prime_editing_indel_freq	16.7%	40.8%	0.8%	38.3%	+24.1	+37.5
microalgae_biomass_multi_strain	40.0%	32.5%	35.8%	90.0%	–7.5	+54.2
vhh_nanobody_expression_ecoli	0.0%	4.2%	0.0%	2.5%	+4.2	+2.5
chl_yn_cell_line_optimization	64.2%	96.7%	56.7%	94.2%	+32.5	+37.5
cho_antibody_titer	15.0%	8.3%	16.7%	5.8%	–6.7	–10.9
transaminase_amination_conversion	2.5%	0.8%	2.5%	1.7%	–1.7	–0.8
Direction (agn > aware)					3/6	4/6

Domain-agnostic exceeds domain-aware on top-1 match in 3 of 6 tasks under search-OFF (Opus alone) and 4 of 6 under search-ON. Paired Wilcoxon comparing the (domain-agnostic – domain-aware) difference under search-ON to search-OFF returns $p = 0.31$ ($n = 6$), so we cannot reject the null that web-search availability makes no difference at this n . Direction-only, the asymmetry persists or strengthens with search. On microalgae the gap widens (domain-agnostic top-1 match jumps from 33% to 90% with search), while domain-aware proposes the literature-typical reactor type on 57.5% of search-ON iterations. The 6/6 reversal in the main audit (§4.3) is averaged over 8 models; Opus alone is 3/6 without search and 4/6 with search.

Cross-figure extension: feedback-actionable subset. As a second sample, we re-ran the same protocol on 5 biology tasks where Claude Opus 4.7’s domain-aware iter-30 LLM-best-result match rate is worst on Figure 4’s feedback-actionable subset. Two of the five tasks (`hek293_prime_editing_indel_freq`, `vhh_nanobody_expression_ecoli`) overlap with the literature-divergent set above and serve as cross-validation. Total: 5 tasks $\times$ 2 conditions $\times$ 4 runs $\times$ 30 iterations = 1,200 LLM calls (1,188 web-search calls logged across the run). Per-task per-condition top-1 match rates use the same key-categorical single-match definition as Table 4.

Domain-agnostic remains $\geq$ domain-aware on 4 of 5 tasks under both search-OFF and search-ON; the lone reversal is `adcp_target_phagocytosis`, where Opus’s domain-aware proposes the published-best antibody isotype (IgG1) on most iterations even without search and search inflates that further (70.0% $\rightarrow$ 87.5%). On the four other tasks the domain-aware match rate stays at or near zero whether search is on or off, while domain-agnostic shows a small positive lift on `mab` (+3.3 percentage points absolute under search-ON). Two tasks (`e_coli_inducible_gfp_yield`, `vhh_nanobody_expression_ecoli`) show 0% domain-aware match on both conditions, indicating that for these tasks Opus’s prior simply does not propose the published-best key value

Table 5. Per-iteration published-best key-categorical match rate on the 5 feedback-actionable tasks under Claude Opus 4.7 only, search-OFF (canonical) vs. search-ON (forced). Same definition as Table 4. Tasks marked [†] overlap with the literature-divergent set; deviations from Table 4 on those rows reflect run-to-run sampling variability of Opus rather than a protocol change.

Task	D-aware off	D-agnostic off	D-aware on	D-agnostic on	(agn – aware) off	(agn – aware) on
adcp_target_phagocytosis	70.0%	30.8%	87.5%	71.7%	−39.2	−15.8
mab_developability_aggregation	0.0%	2.5%	0.0%	5.8%	+2.5	+5.8
e_coli_inducible_gfp_yield	0.0%	0.0%	0.0%	0.0%	+0.0	+0.0
hek293_prime_editing_indel_freq [†]	16.7%	40.8%	10.8%	15.0%	+24.2	+4.2
vhh_nanobody_expression_ecoli [†]	0.0%	4.2%	0.0%	4.2%	+4.2	+4.2
Direction (agn > aware)					4/5	4/5

at all, even with web evidence available. **Overlap with Table 4.** The vhh row reproduces the prior search-ON numbers exactly (0.0% / 4.2% vs. 0.0% / 2.5%; within 1.7 percentage points). The hek293 row’s domain-agnostic search-ON rate falls from 38.3% in Table 4 to 15.0% here, a −23.3 percentage point deviation that exceeds our ± 5 percentage point pre-specified threshold. Both rows used identical task lists, prompts, model ID, and search configuration, so the gap reflects run-to-run sampling variability of Claude Opus 4.7 (the model is non-deterministic and we do not seed). The cross-figure direction-only conclusion (4/5 vs. 4/6) is unchanged.

A.22 Oracle-proximity diagnostic

We test whether the §4.3 domain-aware vs. domain-agnostic gap on biology can be reduced to an oracle-proximity confound, i.e., whether domain-aware LLMs win bsf-AUC by staying close to the training set where the oracle is best calibrated.

Method. For each (bio task, model, condition, run, iteration), we compute the Euclidean distance (d_1) from the LLM-proposed parameter vector to its 1-nearest neighbor in the per-task literature dataset. Numeric parameters are z-scored on the training distribution; categorical parameters are dropped from the distance metric (a first-pass simplification). NaN-imputation: missing numeric fields use the training mean (zero in z-space). We compute (i) the median d_1 per (model, condition) cell and (ii) the per-task Spearman rank correlation between d_1 and oracle prediction across all proposed points.

Result 1: Domain-aware proposals lie systematically closer to training. Median d_1 across all bio (model, condition) pairs is 1.47 z-units under domain-aware vs. 2.71 under domain-agnostic; paired-by-(model, task) Wilcoxon gives $p \approx 1 \times 10^{-12}$ ($n = 275$ pairs). This holds across all 8 models. The proposal-side mechanism predicted by the oracle-proximity hypothesis is confirmed.

Result 2: Closer-to-training does not yield higher oracle scores in aggregate. Pooled across 64,739 bio (task, model, condition, run, iter) points, the Spearman correlation between d_1 and the oracle prediction is +0.221, task-bootstrap 95% CI [+0.047, +0.384]. The sign is opposite of the proximity-confound prediction: farther-from-training points score *slightly higher* on average. Per-task: 47% of bio tasks show positive Spearman (farther pays); 53% show negative (closer pays); the heterogeneity is large enough to flip sign on a substantial subset, indicating no unifying effect.

Result 3: The score-vs-density relation is task-specific. Five biology tasks show strong negative Spearman ($r \leq -0.40$, meaning closer-to-training pays substantially): chl_yn_cell_line_optimization

high_five_protein_yield, qpcr_efficiency, transaminase_amination_conversion, sf9_protein_yield. Five show strong positive Spearman ($r \geq +0.50$, meaning farther pays): baculovirus_titer_sf9, cho_k1_antibody_titer, mab_developability_aggregation, s_aureus_biofilm_inhibit_perc, fcgr_binding_affinity.

Implication. Domain-aware’s tendency to hug the literature prior is a genuine proposal-side bias but not a sufficient mechanism to explain the bsf-AUC gap on biology. Closer proposals do not systematically receive higher oracle scores. The §4.3 cross-subject pattern is therefore a mix of oracle-alignment effects (domain-specific) and real LLM-prior misalignment (anchored in published wet-lab outcomes by §4.3’s 6 documented literature-divergent cases). Limitations: (i) categoricals are dropped from the distance metric, so categorical-only divergences are invisible; (ii) the chlamydomonas_lipid_yield_comprehensive task contributes extreme out-of-bounds proposals that inflate means but not medians, which we report throughout.

A.23 Oracle models

Each task’s oracle is a supervised model trained on published experimental data using leave-one-out cross-validation for model selection. We consider gradient boosting, random forest, ridge regression, SVR, and stacked ensembles. The model with the highest LOO R^2 is selected automatically. Biology oracles have median LOO $R^2 = 0.89$ (range 0.38–0.99, $n = 45$). Education oracles have median LOO $R^2 = 0.32$ (range 0.002–1.00, $n = 10$). The lower R^2 on education tasks reflects the genuinely noisier nature of human learning outcomes, a property of the underlying scientific problem, not a benchmark artifact. We retain low- R^2 education tasks deliberately, since filtering them would bias the benchmark toward only “clean” problems and away from realistic conditions in social science. Oracle predictions are deterministic given a parameter vector.

A.24 GP-UCB baseline

For each task, we run 1,000 independent GP-UCB optimizations (Gaussian process with Matérn 5/2 kernel, UCB acquisition with $\beta = 2.0$, 100 random candidates per acquisition step, 1 random seed point) over 30 iterations to establish a strong statistical baseline. **GP-UCB queries the same oracle as the LLM conditions:** at each iteration the selected candidate is scored by the task’s oracle model, exactly the same `oracle_predict` function the LLMs receive feedback from. GP-UCB operates on the parameter space encoded as $[0, 1]$ -normalized numeric features plus one-hot categoricals, with no semantic labels, no domain names, and no parameter units. I.e., it receives the same information environment as the domain-agnostic LLM condition. We report the median bsf-AUC across runs. LLM performance is GP-normalized: $\text{bsf-AUC}_{\text{norm}} = (\text{bsf-AUC}_{\text{LLM}} - \text{bsf-AUC}_{\text{GP}}) / \max(|\text{bsf-AUC}_{\text{GP}}|, \epsilon)$ with $\epsilon = 0.01$, so 0 indicates parity with the GP baseline. The floor only activates when GP’s bsf-AUC is within 1% of zero, which does not occur on any task in our panel. We report median GP-normalized bsf-AUC (Figure 13) alongside the outperformance view as a robustness check against near-zero denominators. Because GP-UCB shares the oracle and parameter space with both LLM conditions, any artifact in the oracle (e.g., inflated scores near literature-typical designs) is inherited by the GP baseline, not removed by normalization. See §5 for how this lets us rule out simple oracle-smoothness confounds.

200-run GP consistency check. Main figures use a 4-run-matched GP baseline (median of 4 GP-

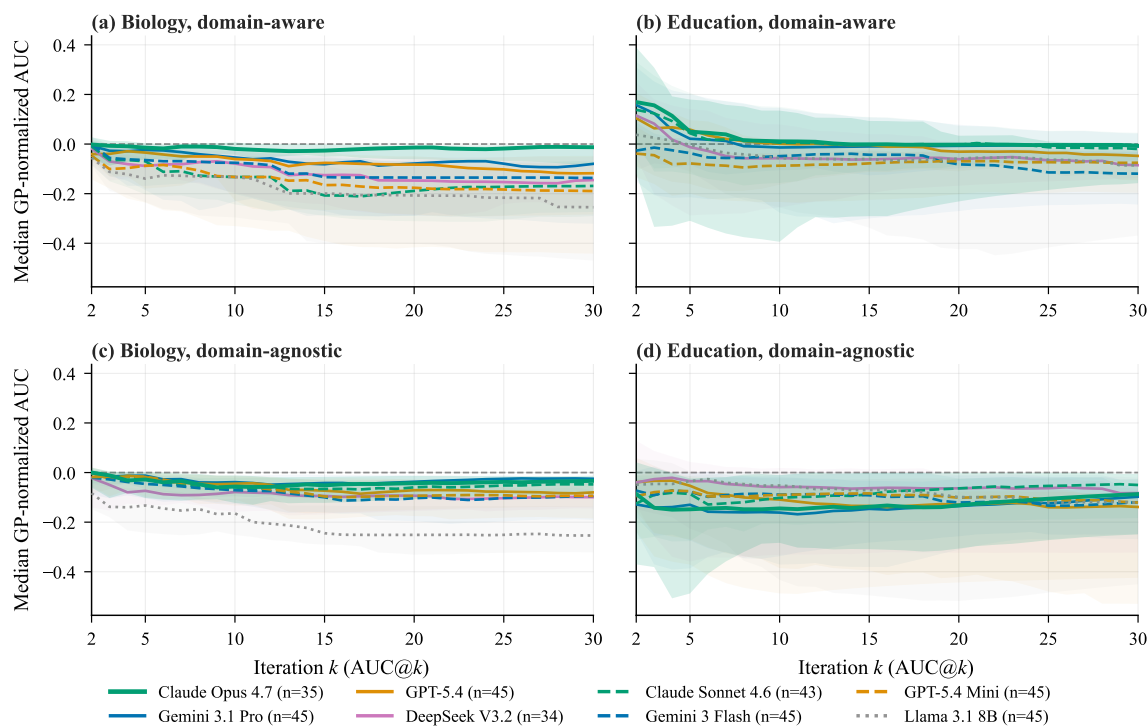


Figure 13. Per-iteration median GP-normalized bsf-AUC across domain and condition. Top row: domain-aware. Bottom row: domain-agnostic. Left column: biology (45 tasks). Right column: education (10 tasks). Zero is GP parity. Negative means LLM below GP. Shaded bands are bootstrap 95% CIs on the median.

UCB runs per task, matching the LLM 4-run protocol). As a consistency check we recomputed per-model outperformance rates against the canonical 200-run GP `cum_median` baseline. Numbers shift by ≤ 7 percentage points and model ordering is preserved. Under `cum_median`, domain-agnostic biology outperformance: Pro 62%, Opus 50%, GPT 44%, Sonnet 40%, Flash 38%, DeepSeek 38%, Mini 31%, Llama 11%; domain-aware: Opus 64%, DeepSeek 46%, Pro 44%, Sonnet 37%, GPT 33%, Flash 29%, Mini 24%, Llama 16%. No model crosses 50% with a strictly-above CI under either baseline.

A.25 Domain-agnostic outperformance vs GP-UCB: prompt manipulation moves performance within the GP-parity band, not across it

Figure 3b in the main text shows per-model outperformance vs. GP-UCB on biology under both prompt conditions. Figure 14 expands the domain-agnostic view. Most frontier models cluster near 50% under both conditions; no model’s 95% CI is strictly above the null. Switching from domain-aware to domain-agnostic shifts outperformance by mixed amounts across models (Sonnet and Gemini 3 Flash gain notably under domain-agnostic, Opus 4.7 declines slightly), but the prompt manipulation moves performance within the GP-parity band rather than across it. Stripping prompt semantics does not unlock latent optimization ability above GP-UCB. The `bsf-Outcome@30` contrast is much smaller.

Implication. Same models, same oracle, same parameter space, only the prompt’s semantic content changes. Outperformance doubles or more for several models. This is consistent with the

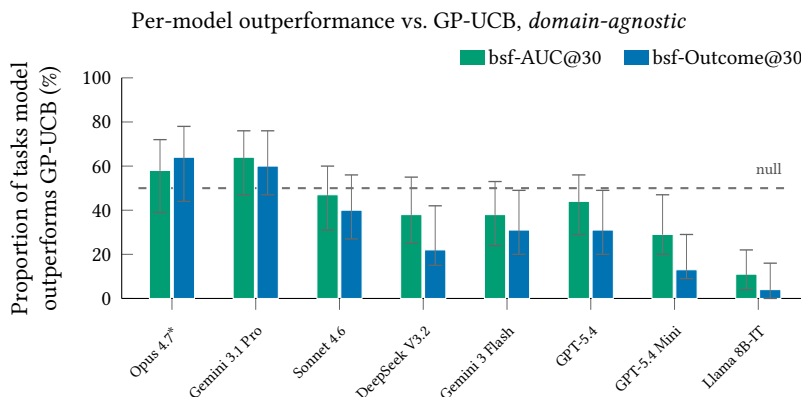


Figure 14. Per-model outperformance vs. GP-UCB on biology under the *domain-agnostic* condition. Mirror of Figure 3b, which shows the domain-aware condition. Dashed line marks the 50% null. Error bars are 2-level bootstrap 95% CIs (4-run-matched GP). No model’s CI is strictly above the 50% null. Domain-agnostic differs from domain-aware in mixed directions across models. The bsf-Outcome@30 contrast is small. *Opus’s coverage is 36/45 biology tasks (§A.8).

priors-vs-feedback weighting noted in §4.3. With domain semantics in the prompt, the literature prior tends to override oracle feedback. Stripping the semantic content lets models use feedback better, sometimes outperforming GP-UCB.

A.26 Education companion: per-model GP-UCB outperformance, both prompt conditions

Companion to Figure 3a (biology) on the 10-task education panel. Frontier models (Opus 4.7, Gemini 3.1 Pro, Sonnet 4.6) clear the 50% null at iter 30 under domain-aware, with point estimates near 90% under bsf-AUC@30 (Figure 15). Switching to domain-agnostic moves frontier models down toward parity with GP-UCB, opposite to the biology pattern in Figure 14 where domain-agnostic helps several models. The cross-subject sign flip described in §4.3 is visible at the per-model bar level in this companion.

A.27 Per-iteration outperformance vs. GP-UCB under the domain-agnostic condition

Figure 16 shows the per-iteration outperformance version of Figure 14, which reports iter-30 only. It mirrors the per-model trajectory view under the domain-agnostic condition. Most frontier models cluster near 50% at iter 30 under domain-agnostic. The domain-aware vs. domain-agnostic comparison is mixed across models. Some gain outperformance under domain-agnostic (Sonnet, Gemini 3 Flash) while others stay similar or decline slightly (Opus 4.7).

A.28 Modal-rank distribution conditional on missing the best-result

For each (model, run) trajectory on the 6 literature-divergent biology tasks, we identify the categorical value the trajectory proposes most often (its mode) and look up that value’s rank in the published literature dataset’s frequency distribution (rank 1 = literature-typical = most-common value in the published literature dataset, rank 2 = second-most-common, etc.). Conditional on the trajectory missing the best-result, we tabulate where this most-frequent proposed value falls.

When domain-aware misses, its modal value sits at shallower frequency ranks than domain-

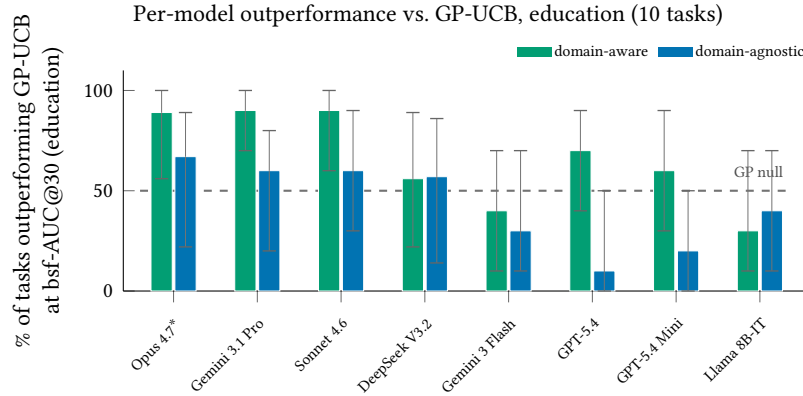


Figure 15. Per-model bsf-AUC@30 outperformance vs. GP-UCB on the 10 education tasks, both prompt conditions. Companion to Figure 3a (biology) and Figure 14 (biology, domain-agnostic). Dashed line: 50% null. Error bars: 2-level bootstrap 95% CIs (4-run-matched GP). Frontier models clear the null under domain-aware. Switching to domain-agnostic moves frontier models toward parity, the opposite direction from biology, consistent with the cross-subject reversal in §4.3. *Opus coverage is 9 / 10 education tasks.

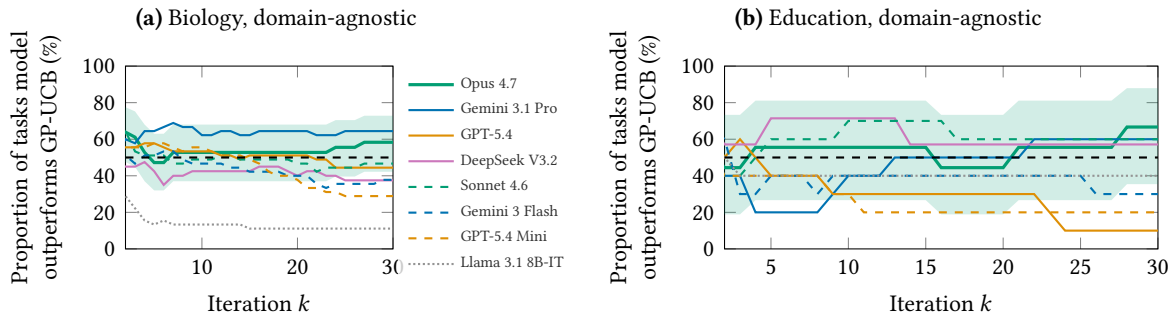


Figure 16. Per-iteration pass rate vs. GP-UCB under the domain-agnostic condition. (a) Biology (45 tasks). (b) Education (10 tasks). Dashed line marks the 50% null; shaded band shows Wilson 95% CI for the highlighted model (Opus 4.7). Compare to Figure 3b (domain-aware bar version) and §A.12.

Table 6. Modal-categorical rank distribution conditional on missing the best-result, on the 6 literature-divergent biology tasks. Rank 1 = literature-typical, rank 4+ = rare values in the published literature dataset. GP-UCB is run with one seed point and 29 acquisition steps per run.

Optimizer	n misses	Mean modal rank	Rank 1 (literature-typical)	Rank 2–3	Rank 4+
Domain-aware	112	2.89	31.2%	50.0%	18.8%
Domain-agnostic	113	3.53	17.7%	54.9%	27.4%
GP-UCB	17	3.12	23.5%	47.1%	29.4%

agnostic’s misses: 31.2% on the literature-typical value itself versus 17.7%, and only 18.8% on the rare-value tail (rank 4 or worse) versus 27.4%. The mean-rank gap is -0.64 (OLS rank $\sim$ domain-aware indicator with (model, task)-clustered standard errors, $n = 225$ misses across 43 (model, task) pairs; 95% CI $[-1.21, -0.07]$, $p = 0.029$). This is consistent with a “fall back to common options” pattern. We treat it as descriptive support for the audit’s interpretation rather than an independent test, since the per-task paired test is not powered at $n = 6$ and the per-(model, task) pair-level paired test is also not significant ($p = 0.28$, $n = 29$ pairs).

A.29 Task list

The 45 biology tasks (after excluding 1 with oracle LOO $R^2 < 0.3$) span protein engineering (12), cell culture optimization (8), nanoparticle formulation (9), gene editing (6), fermentation (5), and other domains (5). The 10 education tasks cover learning science interventions from published RCTs and quasi-experiments. Per-task statistics for all 55 tasks are listed in Table 7.

Table 7. Per-task statistics for all 55 LEAPBENCH tasks. *dom* = b (biology) or e (education); *obj* = max/min; *n* = number of feature parameters; R^2 = oracle leave-one-out cross-validation R^2 .

Task	dom	obj	<i>n</i>	R^2	Task	dom	obj	<i>n</i>	R^2
adcc_reporter_gene_assay	b	max	8	0.74	microemulsions_particle_size	b	min	5	0.98
adcp_target_phagocytosis	b	max	9	0.96	microemulsions_zeta_potential	b	max	4	0.66
antibody_complement_activation_cdc	b	max	7	0.92	nanoparticle_drug_release_kinetics	b	max	5	0.82
antibody_complement_dependent_cytotoxicity_cdc	b	max	5	0.81	nanoparticle_encapsulation_efficiency_combined	b	max	6	0.90
antibody_construct_expression_yield	b	max	10	0.84	nanoparticle_zeta_potential_optimization	b	max	3	0.95
antibody_expression_stability_cho	b	max	6	0.91	pichia_pastoris_recombinant_protein_yield	b	max	4	0.98
baculovirus_titer_sf9	b	max	4	0.71	polymeric_micelles_drug_loading	b	max	5	0.83
c_glutamicum_lysin_titer	b	max	4	0.95	polymeric_micelles_size	b	min	6	0.77
chl_yn_cell_line_optimization	b	max	5	0.88	polymeric_nanoparticles_zeta_potential	b	max	7	0.94
chlamydomonas_lipid_yield_comprehensive	b	max	6	0.82	qpcr_efficiency	b	max	5	0.79
chlamydomonas_reinhardtii_lipid_extended	b	max	7	0.90	s_aureus_biofilm_inhibit_perc	b	max	6	0.92
cho_antibody_titer	b	max	7	0.93	s_cerevisiae_ethanol_fermentation_yield	b	max	7	0.93
cho_k1_antibody_titer	b	max	6	0.94	sars_cov2_fab_binding_affinity	b	min	3	0.89
e_coli_inducible_gfp_yield	b	max	6	0.93	sf9_protein_yield	b	max	7	0.92
e_coli_mic_amp_activity	b	min	3	0.42	streptomyces_synthesis_antibiotic_yield	b	max	12	0.67
e_coli_recombinant_protein_solubility	b	max	5	0.82	transaminase_amination_conversion	b	max	7	0.59
ecoli_plasmid_dna_yield	b	max	11	0.95	vhh_nanobody_expression_ecoli	b	max	9	0.99
engineered_antibody_stability_tm	b	max	5	0.86	assitments_experiments	e	max	3	0.00
fcgr_binding_affinity	b	min	6	0.88	assitments_mastery_threshold	e	max	3	1.00
fcgr_enhanced_binding	b	min	4	0.91	fractions_sequencing_nc	e	max	3	0.54
hek293_prime_editing_inde1_freq	b	max	8	0.76	interim_testing_categories	e	max	4	0.11
her2_binding_Affinity	b	min	4	0.98	interleaved_retrieval_practice	e	max	4	0.99
high_five_protein_yield	b	max	5	0.94	ottmar_perceptual_cues	e	max	4	0.19
hydrogels_release_kinetics	b	max	6	0.71	oulad_vle_engagement	e	max	5	0.65
lipo_nanoparticle_size_optimization	b	min	4	0.82	pretesting_impossible	e	max	2	0.06
mab_developability_aggregation	b	min	6	0.38	retrieval_practice_wm	e	max	3	0.45
mammalian_crispr_hdr_eff	b	max	6	0.73	star_class_size	e	max	5	0.20
microalgae_biomass_multi_strain	b	max	10	0.89					

A.30 Education task data sources

The 10 education tasks are derived from published RCTs and quasi-experiments. We thank the original authors for releasing their data. Most sources are explicitly licensed for research reuse (CC-BY where indicated below) or obtained from publicly accessible repositories. The exception is `fractions_sequencing_nc`, which is derived from PSLC DataShop (dataset ID 580) and carries a **non-commercial use restriction**; the LEAPBENCH data deposit therefore uses the CC BY-NC 4.0 license uniformly to satisfy that constraint. Oracle models are derived predictors trained on these data, released alongside the benchmark for reproducibility.

- `assitments_experiments`: Prihar et al. (2022), *Exploring Common Trends in Online Educational Experiments*, EDM 2022 (Best Dataset Award). Data: <https://osf.io/59shv/> (CC-BY).
- `assitments_mastery_threshold`: Prihar et al. (2022), same source as above.

- `fractions_sequencing_nc`: Rau, Rummel, Aleven, Pacilio, & Tunc-Pekkan (2017), *Int. J. Science and Mathematics Education*, <https://doi.org/10.1007/s11251-017-9403-7>. Source: PSLC DataShop (dataset ID 580), **non-commercial use only**.
- `interim_testing_categories`: Kang, Ha, & Lee (2023), *Educational Psychology Review* 35:97, <https://doi.org/10.1007/s10648-023-09772-y>. Data: <https://osf.io/mt2e9/>.
- `interleaved_retrieval_practice`: Sana & Yan (2022), *Psychological Science* 33(5), <https://doi.org/10.1177/09567976211057507>. Data: <https://osf.io/aqng6/>.
- `ottmar_perceptual_cues`: Ottmar et al. (2025), *Cognition and Instruction* (Ottmar et al., 2025).
- `oulad_vle_engagement`: Kuzilek, Hlosta, & Zdrahal (2017), *Open University Learning Analytics dataset*, *Scientific Data* 4:170171 (CC-BY).
- `pretesting_impossible`: Seabrooke, Hollins, Kent, Wills, & Mitchell (2021), *Memory & Cognition*.
- `retrieval_practice_wm`: Zheng, Sun, & Liu (2023), *npj Science of Learning*.
- `star_class_size`: Krueger (1999), *Quarterly Journal of Economics* 114(2):497–532. Tennessee STAR project, public-domain administrative data.

B GRPO training details

Data collection. We collect trajectories from all six frontier models running LEAPBENCH on 29 biology training tasks under the **domain-aware** condition only, yielding 5,160 episodes (model-task-run triples with full design + oracle score + rationale sequences over 30 iterations). Mixing all six teachers gives a diverse reasoning-style distribution rather than a single-teacher imitation target. Training on domain-aware trajectories means the model sees the prompt regime where literature priors most strongly activate (§4.3), so bsf-AUC-aligned reward incentivizes overriding those priors when feedback contradicts them. The 11 biology tasks used for held-out evaluation and all 10 education tasks are *never* in the training pool.

Reward. The reward for each trajectory is the cumulative bsf-AUC (mean of the best-so-far learning curve over 30 iterations), then within-group normalized across the 8 trajectories in each GRPO group. This is identical in form to LEAPBENCH’s primary evaluation metric. All 21 evaluation tasks use a uniform maximize objective (see Prompt Design above) so the reward signal has a consistent direction. The three tasks whose conventional objective is minimize (HER2 binding, SARS-CoV-2 Fab binding, FcγR-enhanced binding) are evaluated as maximize tasks here, which corresponds to a different but scientifically valid design goal (e.g., weaker binding for selectivity/safety).

Training. Llama 3.1 8B Instruct with LoRA adapter (Hu et al., 2022) (rank 16, $\alpha=32$, dropout 0.05, applied to q, k, v, o projections). Offline GRPO with KL penalty $\beta=0.1$, group size 8, learning rate 5×10^{-6} , 2 epochs over the fixed trajectory pool. The training curriculum pre-computes advantages from the fixed pool rather than rolling out new trajectories per step, which stabilizes training and makes it cheap enough to run on a single A100. Training loss converges within the 2 epochs. The full run takes roughly 6 hours.

Evaluation. 21 held-out tasks (11 biology + 10 education), 4 independent runs per task, 15 iterations per run (shorter than the main benchmark’s 30 to reduce eval cost). GP-normalization is recomputed at 15 iterations. The trained adapter and training trajectories are released at [release/grpo/](https://github.com/ParetoAI/leap).

Per-task results. Figure 17 shows the full breakdown of GP-normalized Δ bsf-AUC per task.

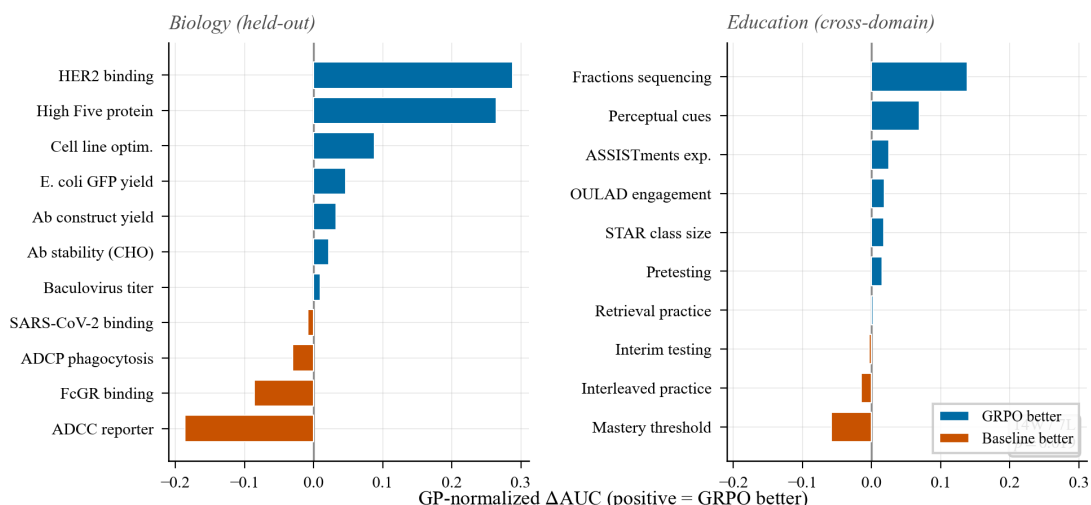


Figure 17. Per-task GP-normalized Δ bsf-AUC across 21 held-out tasks. Biology held-out and education cross-domain (never in training) both show directionally consistent improvement.

Transfer to other trajectory metrics. Training used bsf-AUC-aligned reward, so bsf-AUC improvement is close to in-distribution. To check whether gains transfer to structurally different trajectory metrics, we recompute NIS (number of improving steps) and categorical commitment frequency on the same held-out trajectories. Across all 21 held-out tasks: bsf-AUC improves on 14/21 (Wilcoxon $p = 0.019$, one-sided; $p = 0.038$, two-sided). NIS improves on 11/21 ($p = 0.20$, n.s.). Commitment frequency decreases on only 3/20 tasks where categorical parameters exist ($p = 0.50$, n.s.). The “trainable property” claim is therefore best read as *the bsf-AUC-targeted reward produces bsf-AUC gains that are directionally consistent but transfer only weakly to other trajectory-shape metrics*. We read this as evidence that bsf-AUC-aligned GRPO does not automatically reduce the specific categorical-commitment behavior described in §4.3. A targeted reward (e.g., directly penalizing categorical dominance) would likely be needed to move that metric.

Anchoring on the one literature-divergent task in the eval set. `chl_yn_cell_line_optimization` is the only one of the six §4.3 literature-divergent biology tasks present in the GRPO held-out eval set, so we can directly compare baseline-Llama vs. GRPO-trained-Llama anchoring on it. The literature-typical `feed_rate_strategy` is `dynamic`, and the published-best is `adaptive`. Across 4 runs each, baseline selects the published-best on 3/4 (one run anchored to `dynamic`) and GRPO-trained selects on 4/4 (no run anchored). Sample size is small ($n = 1$ task), so we read this as suggestive that bsf-AUC-aligned GRPO does not degrade anchoring resistance and may modestly improve it on tasks where literature-typical and published-best diverge. A direct targeted-reward test would be needed to confirm this generalizes.

CHO antibody expression trajectory (referenced in §4.4). Figure 18 shows one representative baseline run and one representative GRPO run on the CHO antibody expression task over the first 5 iterations.

Additional trajectory examples. The baseline-vs-GRPO behavioral shift is consistent across held-out tasks. Three examples beyond the E. coli GFP pair described in §4.4:

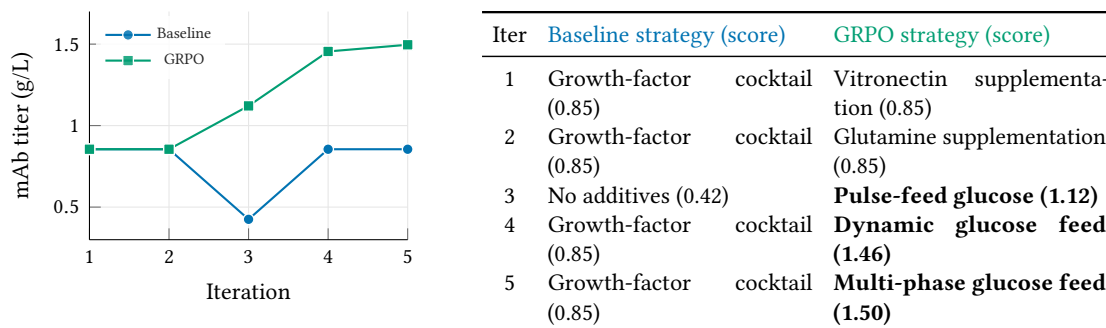


Figure 18. Baseline vs. GRPO on CHO antibody expression, first 5 iterations. One representative run per condition. (a) Per-iteration mAb titer. (b) Strategy per iteration; bold rows mark GRPO’s three successive glucose-feeding refinements. Illustrative example, not a quantitative mechanism claim.

- **Antibody expression (CHO cells).** Baseline tries 5 disconnected strategies (biodiesel additive, cell cycling, temperature control, low glucose, cell line comparison), all scoring 0.85 g/L. GRPO iterates on glucose feeding (high glucose → glutamine → intermittent feed → dynamic rate → multi-phase staging), improving from 0.85 to 1.50 g/L.
- **High Five protein yield.** Baseline achieves 127 $\mu\text{g/mL}$ at iteration 1 but oscillates without sustained improvement. GRPO starts lower (63) but refines cell density and incubation time, converging to 123 $\mu\text{g/mL}$ by iteration 5 with a stable upward trajectory.
- **Baculovirus titer (Sf9 cells).** On the biology-expert-review pair, GRPO climbs monotonically 207M → 487M PFU/mL through iterative MOI and cell-density refinement, while the paired baseline drifts downward 411M → 188M (three biology experts unanimously preferred GRPO).

C Biology-expert review protocol and results

Rationale. The literature-prior stickiness audit (§4.3) and GRPO trainability claim (§4.4) rest partly on quantitative metrics. To check whether these patterns correspond to what a domain practitioner would independently identify, we conducted a small-scale structured review with $N=3$ biology experts. This is a small sample and not a statistical validation. Its purpose is to provide supporting qualitative evidence.

Participants. Three biology experts with experimental-biology backgrounds. Biology experts were blinded to system identity (baseline vs. GRPO; model names in Part A), to our hypotheses, and to condition labels. They were compensated for their time at a rate well above minimum wage, consistent with the NeurIPS Code of Ethics.

Materials. Five paired trajectories (baseline vs. GRPO-trained on held-out tasks: CHO antibody stability, baculovirus titer, E. coli GFP yield, ADCP phagocytosis, perceptual cues) and four individual trajectories (benchmark models under domain-aware condition: Gemini 3 Flash on CDC lysis, GPT-5.4 Mini on CHO stability, Claude Sonnet 4.6 on E. coli GFP, Gemini 3.1 Pro on prime editing). Each trajectory shows 5 iterations of proposed designs, rationales, and oracle scores. Trajectories were curated to span the anchored/responsive range (instrument labels) rather than randomly sampled. Selection was not pre-registered.

Protocol. Biology experts were shown a written definition of anchoring (“persisting with

similar or domain-typical designs despite weak or stagnant outcomes”) and instructed to focus on whether later proposals reflect adjustment to earlier results, not final score. The review had three parts:

- **Part B (pairwise GRPO comparisons).** For each of 5 baseline-vs-GRPO trajectory pairs (system identity hidden, assignment randomized per pair), biology experts were asked which of two unlabeled trajectories (A or B) showed stronger iterative reasoning and why. Outcome scores were visible but the prompt framed the question around trajectory quality, not final score.
- **Part A (individual classifications + quality ratings).** For each of 4 individual benchmark-model trajectories (all domain-aware), biology experts classified the trajectory as *anchored* (persisting with similar or domain-typical designs despite weak outcomes) or *responsive* (adjusting proposals in response to earlier results), and rated reasoning quality on a 1–5 scale (1 = generic, 5 = expert-level).
- **Part C (free-response reflection).** Three open questions: (1) is anchoring a recognizable pattern in experimental science, and did you see it in these trajectories. (2) which trajectories showed the strongest and weakest reasoning, and why. (3) were there any other patterns in the trajectories worth flagging.

Part B was presented first, Part A second, Part C last. This ordering avoids priming from the explicit “anchored/responsive” classification before the blinded pairwise task. Estimated time: 45 minutes.

Part B results, pairwise preferences.

Table 8. Part B pairwise preferences. $N=3$ biology experts; one did not answer Pair 5. System-to-condition assignment was randomized per pair; biology experts were blinded.

Pair	Baseline identity	GRPO preferred
CHO Antibody Stability	A=Baseline, B=GRPO	3/3
Baculovirus Titer	A=GRPO, B=Baseline	3/3
E. coli GFP Yield	A=Baseline, B=GRPO	1/3
ADCP Phagocytosis	A=GRPO, B=Baseline	3/3
Perceptual Cues	A=Baseline, B=GRPO	2/2
Total		12/14 (86%)

Part A results, classifications and quality ratings.

Table 9. Part A classifications and quality ratings. $N=3$ biology experts. Quality scale: 1 = generic, 5 = expert-level.

Trajectory	Expected	Classifications ($N=3$)	Quality mean	Quality values
CDC Lysis (Flash)	Anchored	3 Anchored / 0 Responsive	3.33	3, 3, 4
CHO Stability (Mini)	Anchored	2 Anchored / 1 Responsive	3.00	3, 2, 4
E. coli GFP (Claude)	Responsive	3 Responsive / 0 Anchored	4.33	5, 4, 4
Prime Editing (Pro)	Responsive	2 Responsive / 1 Anchored	4.00	5, 4, 3

Across the four trajectories, biology experts agreed with our labels on 10 of 12 individual

classifications (83%). Anchored trajectories received a mean quality rating of 3.17 versus 4.17 for responsive trajectories, a separation in the same direction as our labels (Mann–Whitney U , $p = 0.031$, one-sided).

Part C results, free-response themes ($N=3$).

On anchoring as a recognizable pattern: Expert 1 wrote, “Anchoring can be a recognizable pattern ... researchers will often make incremental adjustments within a familiar design space ... [which] could lead to persistent optimization of suboptimal conditions if the results are not critically reevaluated.” Expert 2 wrote, “A couple did show quite firm anchoring, persisting with changing one tiny thing that clearly wasn’t working.” Expert 3 wrote, “In usual practice, scientist tries to change the experimental design if there is negative feedback ... in some cases, there could be specific mechanism behind persisting with familiar design.”

On strongest/weakest reasoning: Biology experts converged on CHO Antibody Stability GRPO and E. coli GFP Yield as the strongest trajectories (identifying an effective feeding strategy and iterating on it; discovering the thyA-supplementation mechanism), and on ADCP Phagocytosis baseline as one of the weakest (“repeatedly used the same IgG1-based design and nearly identical rationale across all steps”).

On reasoning depth: Biology experts noted variation in mechanistic grounding across trajectories. Expert 2 added that many trajectories “change too many things at once - you’ll never know which change was actually having the effect”, highlighting a DoE-style concern separate from the literature-prior stickiness construct.

Interrater agreement. On the Part A binary classification: overall agreement 10 of 12 (83%). Pairwise: Expert 1 vs 2 = 4 of 4 (100%). Expert 1 vs 3 = 2 of 4. Expert 2 vs 3 = 2 of 4 (Expert 3 disagreed with both on Trajectories 2 and 4). Fleiss’ $\kappa = 0.33$ across the three raters (fair agreement, supporting evidence, not confirmatory). Reproducible via `expert_review/analyze.py`. On Part A quality ratings, the responsive vs. anchored separation is consistent across experts (Experts 1 and 2: responsive $\geq$ anchored on every pair; Expert 3: mixed).

Known limitations. (i) Small sample ($N=3$ biology experts). Not a statistical validation of the literature-prior stickiness construct. (ii) Anchoring was defined in the instructions rather than discovered by biology experts. Independent identification evidence comes only from Part C free-response content. (iii) Trajectories were curated, not randomly sampled. Part A trajectories were selected to span the anchored/responsive range. Part B paired trajectories were selected because GRPO appeared to learn better than baseline. Selection was not pre-registered. (iv) Trajectories shown at 5 iterations rather than the 30-iteration benchmark length. A trajectory that would eventually escape anchoring is indistinguishable from one that would not at 5 iterations. (v) The instrument primes the anchoring construct before Part C free-response questions, weakening claims about independent identification. However, Part C questions 1–2 are phrased without reference to the construct. (vi) One pilot biology expert flagged contaminated rationale text in two trajectories (Baculovirus Steps 3–5 and ADCP Steps 1–5). These were formatting errors corrected before subsequent biology-expert sessions (the parameters and scores shown were always correct).

C.1 Trajectories shown to reviewers

This subsection reproduces the nine trajectories that biology experts saw during Part B (five paired comparisons, baseline vs. GRPO) and Part A (four individual benchmark-model trajectories, domain-aware condition). System identity and condition labels were blinded from reviewers at review time. The full protocol, including the anchoring definition shown to reviewers, is in §C.

Part B: paired comparisons.. Pair 1: CHO Antibody Stability. Maximize monoclonal antibody titer (g/L) in CHO cells.

System A:

Step	Media	Additives	Cell line	Rationale	→ Outcome
1	CDM with high glucose	glycerol biodiesel additive	CHO-S	Adding biodiesel derivative may enhance cellular lipid metabolism and subsequently increase titer.	0.85
2	CDM with low glucose	cell wall-acting proteins and antifoaming agents	CHO-K1	Minimizing bioreactor volume may reduce cellular byproducts while increasing productivity for high-cell-density culture; optimizing cell density by using regulatory agents can foster improved monoclonal antibody production.	0.85
3	CDD with low glucose	antifoaming agents and temperature shift from 37°C to 33°C at day 4	CHO-K1	Precise temperature control during temperature-regulated fed-batch operation could allow hyper-feed to promote sustained cell growth while ensuring that metabolic pathways can support enhanced monoclonal antibody expression levels.	0.85
4	CDM with extreme low glucose and high CO2	antifoaming agents	CHO-K1	As cell lines are likely at their saturation point for monoclonal antibody production, pushing the conditions to extreme could potentially reveal upper limits of productivity or indicate where productivity may start to degrade.	0.85
5	CDM with low glucose	antifoaming agents	CHO-DG44	Comparing the performance of CHO-DG44 and CHO-K1 cell lines may reveal inherent differences in their ability to produce monoclonal antibodies and identify new opportunities for optimization.	0.85

System B:

Step	Media	Additives	Cell line	Rationale	→ Outcome
1	Hydro-Opti	Vitronectin	CHOK1-Lectin	Providing high glucose and vitronectin will promote efficient cell growth and antibody production	0.85
2	CD CHO	Glutamine-rich supplement	CHO-K1	Providing additional glutamine will satisfy the energy and biosynthetic requirements of CHO cells, potentially increasing antibody secretion	0.85
3	Hydro-Opti	Pulse-feed high glucose	CHO-K1	Testing the effect of intermittent high glucose feeding on prolonging cellular metabolic capabilities and increasing antibody production	1.12
4	Hydro-Opti	Dynamically adjust glucose feed rate based on cellular growth rate	CHOK1-Lectin	Testing the effect of real-time glucose adjustment on maintaining optimal biosynthetic processes and antibody production	1.46
5	Hydro-Opti	Feeding glucose in multiple stages during cultivation	CHOK1-Lectin	Evaluating the effect of staged glucose provision on cell metabolic adaptation and consecutive antibody production	1.50

Pair 2: Baculovirus Titer. Maximize baculovirus titer (PFU/mL) in Sf9 cells.

System A:

Step	Cell density	MOI	Incub. time	Rationale	→ Outcome
1	1250000	4	96	Higher MOI increases viral replication rates, potentially leading to higher titer yields.	207M
2	1650000	5.5	100	This hypothesis tests if higher MOI values lead to diminishing returns or even decreased titer yields due to viral saturation.	261M
3	2150000	2.2	118	This hypothesis tests the claim that higher cell densities and extended culture durations improve titer production by optimizing viral replication and expression efficiency.	387M
4	3200000	1.2	126	This hypothesis tests the claim that extremely high cell densities ($> 3.2 \times 10^6$ cells/mL) can lead to even higher titer yields by pushing the limits of viral replication and expression efficiency.	455M
5	3800000	1.5	130	This hypothesis tests if further increasing cell density beyond the current limit will lead to a saturation point where titer production levels off or declines.	487M

System B:

Step	Cell density	MOI	Incub. time	Rationale	→ Outcome
1	2450000	1.5	96	Higher cell density and MOI at early time point may enhance virus production.	411M
2	1900000	1.2	90	Testing a middle density (allowing for efficient virus production) combined with a short incubation time to maximize titers.	377M
3	2200000	1	100	Testing the optimal MOI and incubation time at a moderate cell density of 2.2 million cells/mL to maximize titer, given the observed negative effect of high cell density.	395M
4	1500000	1	200	Testing the effect of doubling incubation time at a lower cell density to see if titer can be further improved due to reduced cell-related limitations.	250M
5	1000000	0.5	220	Testing the effect of lowering MOI and increasing incubation time in conjunction with a lower cell density to maximize virion production.	188M

Pair 3: E. coli GFP Yield. Maximize GFP fluorescence yield (RFU) in E. coli.

System A:

Step	Strain	Promoter	Gene mod	Rationale	→ Outcome
1	E. coli BL21(DE3)	OXB20	WT	Optimizing the promoter type for stronger expression may yield higher GFP production.	160
2	E. coli W3110	OXB20	WT	Will the OXB20 promoter still yield high GFP production in E. coli W3110?	169
3	E. coli C44(DE3)	OXB20	WT	Will OXB20 promoter still yield high GFP production with a different strain, E. coli C44(DE3)?	160
4	E. coli BL21(DE3)	OXB20	crgA	Will modifying the plasmid with crgA further optimize GFP production in E. coli BL21(DE3) using OXB20 promoter and BHI medium?	160
5	E. coli BL21(DE3)	OXB20	crgA + nusG + rrnD	Will adding rrnD gene modification to the existing crgA and nusG modifications result in higher GFP production in E.	481

System B:

Step	Strain	Promoter	Gene mod	Rationale	→ Outcome
1	E. BL21(DE3)	coli T7	WT	The T7 promoter is a strong and commonly used promoter in E.	174
2	E. BL21(DE3)	coli T7	0.09 GFP_cer_placIQ_pbla-thyA	- Testing the effect of supplemental thymidine on T7 promoter expression in a high copy number E.	45,936
3	E. BL21(DE3)	coli T7	0.09 GFP_cer_placIQ_pbla-thyA	- Testing the effect of using a p15A ori plasmid with supplemental thymidine on T7 promoter expression in an E.	45,936
4	E. BL21(DE3)	coli T7	0.09 GFP_cer_placIQ_pbla-thyA	- Testing whether adding a p-lacI inducer (IPTG) to the high-copy number T7 promoter configuration can further enhance GFP yield in E. coli.	45,936
5	E. BL21(DE3)	coli T7-G10	0.09 GFP_cer_placIQ_pbla-thyA	- Testing whether the T7-G10 promoter, which is an even stronger variant of the T7 promoter, can still achieve high GFP yield in combination with high-copy number plasmid and IPTG in E. coli.	45,936

Pair 4: ADCP Phagocytosis. Maximize antibody-dependent cellular phagocytosis (%).

System A:

Step	Isotype	Fc mutation	Format	Rationale	→ Outcome
1	IgG3	E430G	monoclonal	Combining IgG3 isotype with glycoengineered antibody results and optimized Fc region mutations may significantly enhance the engagement of activating FcγR on phagocytes.	18.75
2	IgG1	2h-AEA	HexaBody	Combining the activating Fc mutation, 2h-AEA, with a HexaBody format antibody, which has multiple antigen binding sites to effectively increase the number of antibodies binding to the target cell surface, should significantly increase Fc receptor engagement and enhance phagocytic activity.	25.77
3	IgG3	E430G	combination RTX-IgG2	with Combining the activating Fc mutation, E430G, with the addition of an RTX-based antibody, RTX-IgG2, which is known to induce Fc receptor shedding, could synergistically enhance Fc receptor engagement and phagocytic activity.	16.84
4	IgG2	2h-DAA	monoclonal	Combining the activating Fc mutation, 2h-DAA, with IgG2 isotype, monoclonal antibody format, and glycoengineered glycan composition to effectively increase Fc receptor engagement and enhance phagocytic activity.	24.62
5	IgE	silent	HexaBody	Using an IgE isotype with Fc region silenced, and combining it with a HexaBody format that increases the number of antibodies interacting with Fc receptors, and an activating Fc mutation, 2h-AEA, may unexpectedly enhance phagocytic activity despite the generally low affinity of IgE antibodies for Fc receptors.	35.69

System B:

Step	Isotype	Fc mutation	Format	Rationale	→ Outcome
1	IgG1	none	monoclonal	This hypothesis leverages a well-characterized IgG1 monoclonal antibody format with standard conditions to establish a baseline for phagocytic activity.	44.62
2	IgG1	none	HexaBody	This hypothesis tests whether glycan engineering to enhance Fc receptor binding improves phagocytic activity in a HexaBody format.	45.49
3	IgG1	none	monoclonal	This hypothesis tests whether limiting the effector-to-target ratio can significantly improve phagocytic activity by reducing competition for target cells.	46.09
4	IgG1	none	monoclonal	This hypothesis tests whether glycan engineering enhances phagocytic activity, and whether the combination of glycan engineering and reduced E:T ratio is additive.	40.19
5	IgG1	2h-DAA	monoclonal	This hypothesis tests whether enhancing Fc receptor binding through 2h-DAA mutation combined with glycoengineered glycan composition improves ADCP.	40.19

Pair 5: Perceptual Cues (Education). Maximize student accuracy on order-of-operations problems.

System A:

Step	Spacing	Color	# problems	Rationale	→ Outcome
1	congruent	congruent	50	Higher practice and consistent perceptual cues support accurate order-of-operations performance, even with slower response times.	0.77
2	none	none	20	Student accuracy on order-of-operations problems is more strongly predicted by speed of practice (faster is better) than by perceptual cues (congruent or incongruent are similar).	0.04
3	congruent	congruent	20	Congruent perceptual cues will improve accuracy for students who practice at slower speeds (as indicated by higher response times), suggesting that cue type and practice speed interact in their effects.	0.38
4	incongruent	incongruent	40	Inconsistent perceptual cues may actually improve accuracy for students who practice at faster speeds (as indicated by lower response times), potentially by encouraging closer attention to specific details within the problems.	0.24
5	congruent	congruent	30	Congruent perceptual cues will result in higher accuracy when students have reached a moderate level of practice proficiency (as indicated by a response time that is not extremely slow or fast) and are able to apply this knowledge effectively.	0.26

System B:

Step	Spacing	Color	# problems	Rationale	→ Outcome
1	none	congruent	15	Provision of congruent color cues can aid students with minimal spacing cues and moderate practice effort.	0.58
2	congruent	none	49	Does explicit spacing cues foster accuracy in high volume practice settings without additional color cues?	0.65
3	congruent	congruent	19	Does the concurrent use of explicit spacing and color cues improve accuracy compared to using just one or no explicit cues in a moderate volume practice setting?	0.75
4	congruent	none	15	Does exposure to explicit spacing cues followed by explicit color cues improve accuracy on order-of-operations problems compared to their concurrent use?	0.59
5	none	congruent	45	Does introducing explicit color cues only after achieving higher volume practice significantly improve student accuracy?	0.86

Part A: individual trajectories.. Trajectory 1: Antibody CDC Lysis. Maximize CDC lysis (%) by engineering antibody Fc regions. Baseline: 56.3%.

Step	Fc mutation	Glycosylation	Format	Rationale	→ Outcome
1	E345K	Low fucose	IgG1	The E345K mutation promotes the formation of antibody hexamers on the cell surface, significantly increasing C1q avidity and subsequent complementary mutations in VH and VL domains	26.43
2	E345K/E430G/S440G	Galactosylated	IgG1	Combining multiple hexamerization-enhancing mutations with saturating antigen levels and increased complement availability will synergistically surpass the threshold for C1q recruitment compared to single E345K point mutations.	34.86
3	E345K/E430G/S440G	High terminal galactose	IgG1	High terminal galactosylation increases the conformational flexibility of the CH2 domain, which complements the EFTAE-driven hexamerization to further lower the energy barrier for C1q binding and C4 deposition.	34.86
4	E345K/E430G/S440G	High terminal galactose	IgG1/IgG3 Chimeric	Incorporating IgG3-derived CH2/CH3 sequences into the EFTAE-modified IgG1 backbone will exploit the superior inherent C1q binding of IgG3 while retaining the optimized hexamerization mutations, potentially unlocking a new ceiling for CDC activity.	34.86
5	K326W/E333S/E345K	High terminal galactose	IgG1	Since hexamerization alone has plateaued, adding K326W and E333S will directly increase the intrinsic affinity for the C1q globular head, creating a dual-action mechanism of improved C1q recruitment and structural hexamer stability.	34.86

Trajectory 2: CHO Antibody Stability. Maximize antibody titer (g/L) in CHO cells. Baseline: 1.5 g/L.

Step	Media	Additives	Cell line	Rationale	→ Outcome
1	Chemically defined fed-batch medium	Glucose, glutamine, and essential amino acid feed; mild anti-clumping agent	CHO-K1	Controlled nutrient feeding with reduced glutamine peaks can sustain viability and extend production phase, which often increases monoclonal antibody titer.	0.85
2	Chemically defined high-density fed-batch CHO medium	Glucose, glutamine dipeptide, iron, lipid supplement, and trace elements	CHO-DG44	The pattern I am testing is that higher titers come from cultures that avoid nutrient depletion and metabolic stress while maintaining high viable cell density, so this formulation should improve production by sustaining longer, healthier antibody-secreting phases.	0.85
3	Chemically defined perfusion medium	Glucose, amino acid concentrate, lipid supplement, and trace manganese	CHO-S	The pattern I am testing is that antibody production is limited by intermittent nutrient depletion and waste accumulation, so a stable perfusion-like supply should increase titer by keeping cells in a continuously productive state.	0.85
4	Chemically defined low-lactate fed-batch medium	Galactose, pyruvate, reduced glutamine feed, and trace metals	CHO-K1	The pattern I am testing is that higher monoclonal antibody production is driven by sustained very high viable cell density rather than batch longevity alone, so perfusion with concentrated feeding and salvage-pathway support should increase titer by maximizing cumulative secretion per reactor volume.	0.85
5	Chemically defined production medium with high glucose and balanced amino acids	Sodium bicarbonate, Pluronic F-68, mild osmolality buffer, and amino acid feed	CHO-DG44	The pattern I am testing is that high monoclonal antibody titers arise when pH and osmolality are tightly controlled to reduce cellular stress, so this formulation should improve production by preserving cell viability and secretion capacity under culture stress.	0.85

Trajectory 3: E. coli GFP Yield. Maximize GFP yield (RFU) in E. coli BL21(DE3).

Step	Strain	Promoter	Gene mod	Rationale	→ Outcome
1	E. coli BL21(DE3)	T7	WT	The T7 promoter in BL21(DE3) with a high-copy plasmid and extended 24-hour culture duration should maximize GFP accumulation by leveraging strong T7 RNA polymerase-driven transcription and sufficient time for protein folding and fluorophore maturation.	605
2	E. coli enGenes- eXpress V2 strain BL21(DE3) with Δ thyA genetic modification and reduced repressor levels	T7	0.09 GFP_cer_placIQ_pbla- thyA	- The Δ thyA auxotrophic strain with thymidine supplementation creates a plasmid stabilization selection pressure that maintains high plasmid copy number without antibiotic selection, potentially increasing GFP expression consistency and yield over longer culture durations compared to standard BL21(DE3).	119,572
3	E. coli enGenes- eXpress V2 strain BL21(DE3) with Δ thyA genetic modification and reduced repressor levels	T7	0.09 GFP_cer_placIQ_pbla- thyA	- The dramatic yield increase (119,571 vs 605 RFU) confirms that the Δ thyA auxotrophic selection system is the dominant driver of GFP yield by enforcing plasmid retention without antibiotic cost; I am now testing whether extending culture duration from 36 to 47.5 hours further increases GFP accumulation by allowing more cell growth and protein production under stable plasmid maintenance conditions.	119,681
4	E. coli enGenes- eXpress V2 strain BL21(DE3) with Δ thyA genetic modification and reduced repressor levels	T7	0.09 GFP_cer_placIQ_pbla- thyA	- Since culture duration shows diminishing returns (36h vs 47.5h gave nearly identical yields 119,600 RFU), I am now testing whether switching from high-copy to the pBR322 ori plasmid backbone at maximum culture duration (59.5h) can further increase yield by potentially reducing metabolic burden from plasmid replication while maintaining the critical Δ thyA selection advantage that dominates GFP yield outcomes.	119,681
5	E. coli enGenes- eXpress V2 strain BL21(DE3) with Δ thyA genetic modification and reduced repressor levels	T7 (leaking)	0.09 GFP_cer_placIQ_pbla- thyA	- The results so far show the Δ thyA strain with thymidine supplementation is the overwhelmingly dominant factor (200x yield increase), and culture duration and plasmid ori appear to have plateaued; I am now testing whether a leaky T7 promoter, which allows constitutive low-level expression without requiring inducer addition, combined with the Δ thyA selection system can exceed the current 119,681 RFU ceiling by providing continuous GFP production throughout the entire cul-	120,419

Trajectory 4: Prime Editing Efficiency. Maximize prime editing indel frequency (%) in HEK293T cells.

Step	Cas variant	RTT len	Delivery	Rationale	→ Outcome
1	PEmax	15	plasmid transfection	Combining the PEmax architecture with an epegRNA scaffold in a PE3 setup significantly increases prime editing efficiency through enhanced complex stability and improved reverse transcription.	57.18
2	PE5max	14	plasmid transfection	I hypothesize that using the PE5max protein variant (which incorporates improved mismatch repair evasion) over PEmax, combined with epegRNA and a 14nt RTT, will further maximize efficiency.	65.69
3	PE5max	13	plasmid transfection	I hypothesize that since decreasing the RTT length from 15 to 14 nt improved efficiency with PE5max, further shortening the RTT to 13 nt will maximize efficiency by providing optimal reverse transcription kinetics for the PE3/epegRNA system.	65.78
4	PE5max	13	plasmid transfection	I hypothesize that since decreasing the RTT length provided a modest efficiency gain, increasing the primer binding site (PBS) length to 14 nt will enhance the stability of the hybridization between the epegRNA and the target DNA strand, thereby further increasing prime editing efficiency.	65.78
5	PE5max	13	plasmid transfection	I hypothesize that extending the time post-transfection to 120 hours will yield higher measurable efficiency, as the cells will have more time to complete DNA repair following prime editing without degrading the PE5max machinery.	63.06